UNIFIED MECHANICS

Complete Series

A Unified Physical Theory

Joseph Shields

INTEGRATED OMNIBUS EDITION, VOLUMES I–V

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A CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

Abstract

This integrated omnibus presents Unified Mechanics as one chronological scientific theory: from distinction and retained identity, through carrier geometry and action, to matter, gravitation, cosmology, measurement, and the formal register. It distinguishes exact mathematics, imported results, conditional physical realizations, computations, prospective tests, and localized frontier obligations without averaging them into one vague status. The theory is corrigible statement by statement; its remaining calculations do not demote the architecture they inhabit.

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PART I

Foundations of Logical and Physical Action

CHAPTER 1

The scientific contract

1.1. What this series is

Unified Mechanics is presented here as a *theory of everything* in the precise sense asserted by its corpus: one generative structure is proposed beneath the domains described by general relativity, quantum field theory, and quantum mechanics, and the successful limits of those theories are to be recovered rather than displaced. The corpus specifies a state-and-record grammar, carrier geometry, admissible reduction, action principles, matter representations, cosmological realization, observable maps, and reproducibility rules. Some numerical selectors, constitutive identities, matching conditions, and prospective measurements remain active. Those are localized frontier calculations inside the theory, not grounds for redescribing the theory as absent.

The series therefore asks three questions in sequence:

1. What follows exactly from the framework's stated definitions and postulates?
2. Which parts reproduce or invoke established mathematics and physics?
3. Which additional bridge would be required before an internal construction became a physical explanation or prediction?

This order is more than editorial hygiene. It prevents a familiar error in foundational work: beginning with an evocative physical interpretation, performing correct algebra inside it, and then treating the correctness of the algebra as evidence for the interpretation. Algebra can certify a consequence of a model. It cannot, by itself, certify the model's contact with nature.

> *Every boxed statement carries its epistemic type.* Established result *means a result from the external mathematical or physical literature, used under stated hypotheses.* Internal result *means a consequence of the theory's own definitions or assumptions.* Conditional identification *marks a bridge from an internal object to a physical one.* Conjecture *is a precise open target.* Exposure *records a resolution boundary, non-identifiability, conditional incompatibility, or localized derivational obligation under stated premises.*

1.2. A living edition, not a frozen one

Earlier corpus releases used manifests and printing labels to identify a governing state. Those files remain valuable: they establish provenance, document when a statement entered the corpus, and allow an old calculation to be reproduced. They do not make a scientific conclusion immutable. In this edition, a claim may be corrected, narrowed, superseded, or withdrawn whenever its proof, physical bridge, or evidential basis changes.

The edition uses four records instead of a freeze:

Version history
: identifies the text and date of each public release.

Supersession ledger
: states which claim replaced which earlier wording, and why.

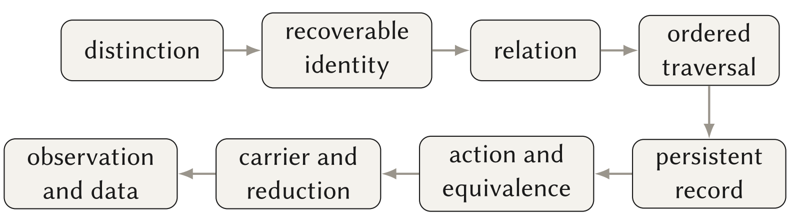
Dependency ledger
: records which definitions and assumptions a result actually uses.

Evidence ledger
: distinguishes derivation, computation, calibration, retrospective comparison, and prospective test.

This policy does not make the series unstable. It makes change explicit. A reader can reconstruct any prior state while treating the most recent scientifically graded statement as the current one.

1.3. The conceptual chronology

The corpus is most intelligible when read as a sequence of increasing physical commitment:



The first four terms are a proposed grammar of system description. A physical record adds dynamics and operational recoverability. An action adds variational or generative structure. A carrier adds a concrete mathematical representation. Observation finally connects the construction to instruments and data. Later volumes follow this same order rather than arranging material by the date or filename in which it happened to be written.

1.4. Authorship and external lineage

The framework and its terminology are Joseph Shields's. Its mathematical neighbourhood is not. Distinction has a particularly close historical contact with Spencer-Brown's calculus of indication (Spencer-Brown, 1969); self-reference and fixed-point arguments belong to a much broader logical tradition, for which Lawvere's categorical diagonal theorem is a useful landmark (Lawvere, 1969). Variational equivalence, quantum probability, causal reconstruction, lattice theory, gauge symmetry, and gravitational action have their own mature literatures. This edition cites those literatures where the corpus uses, parallels, or departs from them.

A citation is not an endorsement. Nor is it a claim that an older theorem anticipated Unified Mechanics. It identifies the exact point of contact, the hypotheses being borrowed, and the boundary beyond which the corpus is proposing something new.

CHAPTER 2

Distinction, identity, and relation

2.1. The primitive vocabulary

The framework begins with three terms.

DEFINITION 2.1 — DISTINCTION

A *distinction* is whatever prevents complete identification of two admissible states or contents. Operationally, a distinction exists relative to a family of tests whenever at least one test yields different records for the two cases.

This operational clause is important. Without a class of tests, "different" is only a label. Let X be a state space and let $\mathscr{M}$ be a family of readout maps $m : X \to Y_m$. Define

$$x \sim_{\mathscr{M}} x' \quad \Longleftrightarrow \quad m(x) = m(x') \quad \text{for every } m \in \mathscr{M}. \tag{2.1}$$

Then the recoverable objects are equivalence classes $[x]_{\mathscr{M}}$, not necessarily microscopic states. This supplies a precise form of the corpus's statement that objecthood is retrospective: an object is what remains identifiable across the transformations and measurements admitted by the problem.

DEFINITION 2.2 — IDENTITY

Identity is recoverability of a distinction under a stated class of admissible transformations. If $\mathscr{T}$ is that class, identity requires that the relevant equivalence class remain discriminable along every $T \in \mathscr{T}$ for which persistence is claimed.

DEFINITION 2.3 — RELATION

A *relation* is an operation or structure through which distinct contents can be compared, coupled, transported, or transformed without being identified merely by participation in the relation.

These definitions are intentionally relational. They do not assert that the world is made only of distinctions, nor that every physical field is reducible to language. They specify the level at which the theory begins.

2.2. Relation to *Laws of Form*

Spencer-Brown begins with the injunction to draw a distinction and develops a calculus in which a mark can function as boundary, indication, crossing, and value (Spencer-Brown, 1969). Unified Mechanics adopts the priority of distinction but refuses that compression at the start. It keeps separate:

1. the act of drawing or producing a difference;
2. the boundary or relation thereby established;
3. the state indicated relative to that boundary;
4. the traversal across or through it;
5. the record by which the traversal remains recoverable.

This is a methodological difference, not a refutation of the calculus of indications. Spencer-Brown studies a compact algebra of forms. The present construction asks what physical machinery is required for a form to be propagated and read. Re-entry can make form behave dynamically; here, dynamics is kept explicit before form is compressed.

2.3. The seven operations

The inherited corpus gives two seven-term lists. The first describes the constitution of a system:

$$\text{distinguish} \prec \text{hold} \prec \text{close} \prec \text{answer} \prec \text{change} \prec \text{order} \prec \text{record}. \tag{2.2}$$

The second describes one traversal through an already articulated system:

$$\text{persist} \prec \text{distinguish} \prec \text{relate} \prec \text{propagate} \prec \text{resolve} \prec \text{return} \prec \text{close}. \tag{2.3}$$

Equation (2.2) is best read as a dependency chain. If those six precedence relations are adopted, the induced finite partial order has a unique linear extension: the displayed order. That is a correct but internal result. The substantive claim lies in the completeness of the seven operations, not in the order-theoretic proof after completeness has been assumed.

Internal result 2.1 — ordering of the proposed grammar

Let $G = \{g_1, \dots, g_7\}$ and impose $g_i \prec g_{i+1}$ for $i = 1, \dots, 6$. The transitive closure is a total order and therefore has exactly one linear extension. This proves the ordering of the stipulated chain; it does not prove that every physical system must instantiate exactly these seven stages.

2.4. Systemhood as controlled unity

The first framework postulate is that a system may retain distinguishable internal contents without ceasing to be treated as one system. Mathematically, unity is then represented not by the absence of parts but by a closure condition: a chosen collection of states and operations remains within a specified domain. This is compatible with many familiar structures—a group closed under composition, an invariant subspace, a Markov process with a conserved state simplex, or a laboratory apparatus with a stable operational boundary. None is selected until further structure is supplied.

The second postulate is stronger: the first distinction does not occur in a preexisting container; its retention supplies the first operational "where." This is an ontological commitment. It may motivate an emergent account of space, but it is not yet geometry. A metric space needs a set, a distance or related structure, and appropriate axioms. A differentiable spacetime needs substantially more. Volume II therefore treats geometry as a later construction, not as a synonym for distinguishability.

EXPOSURE 2.1 — COMPLETENESS OF THE GRAMMAR

The corpus does not yet prove that the two seven-term lists are minimal, independent, or exhaustive. A formal completion requires (i) a syntax of admissible operations, (ii) equivalence rules, (iii) a demonstration that no operation is definable from the others, and (iv) a representation theorem showing that the intended systems instantiate the grammar.

CHAPTER 3

Self-description and the golden fixed point

3.1. The update rule

The first exact numerical construction is generated by a two-component update. Write the state as $\mathbf{v}_n = (a_n, b_n)^\mathsf{T}$, where the first coordinate is the currently resolved share and the second is the held share. The inherited update is

$$\mathbf{v}_{n+1} = M\mathbf{v}_n, \qquad M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \tag{3.1}$$

It adds the two available shares into the next resolved coordinate and carries one clean copy of the present resolved coordinate into reserve. The eigenvalues are

$$\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}, \tag{3.2}$$

and the unique positive dominant eigenvalue is the golden ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}, \qquad \varphi^2 = \varphi + 1. \tag{3.3}$$

INTERNAL RESULT 3.1 — SCALE-STABLE RATIO

For any initial vector with a nonzero component along the dominant eigenvector, repeated application of M, followed by normalization, converges to the positive eigenvector ratio φ. This is the Perron–Frobenius behaviour of the specified positive update matrix. The result selects φ only after M has been chosen.

The corpus sometimes describes this as the fixed point of self-description. Scientifically, the exact statement is narrower: φ is the stable ratio of the particular update in Equation (3.1). Calling that update the unique operation of physical self-description requires a separately proved uniqueness theorem over an explicitly defined admissible class.

3.2. The two-sided normalization

The principal numerical postulate of the corpus is a two-sided reading: a mixed relation is counted both from outward to held and from held to outward. With the inherited normalization this gives

$$r = \frac{1}{2\varphi} = \frac{\varphi - 1}{2} = \frac{\sqrt{5} - 1}{4}, \qquad u = 1 - r = \frac{3 - \sqrt{5}}{4} + \frac{1}{2}. \tag{3.4}$$

Numerically,

$$r \approx 0.3090169944, \qquad u \approx 0.6909830056. \tag{3.5}$$

The factor of two is not a consequence of Equation (3.3). It is the additional assumption A3. Keeping it visible prevents a genuine internal derivation from being misreported as parameter-free physics.

CONDITIONAL IDENTIFICATION 3.1 — RETAINED AND OUTWARD SHARES

If a physical process admits exactly two normalized channels and if its mixed relation is represented by the corpus's two-sided convention, then the channel coordinates are fixed to $r = (2\varphi)^{-1}$ and $u = 1 - r$. Evidence for the physical premise must be supplied separately for each application.

3.3. The three quadratic weights

Squaring the normalized partition yields

$$1 = (u + r)^2 = \underbrace{u^2}_{W_\mathrm{L}} + \underbrace{2ur}_{W_\mathrm{B}} + \underbrace{r^2}_{W_\mathrm{M}}. \tag{3.6}$$

The exact and numerical values are

$$W_\mathrm{L} = u^2 \approx 0.4774575141, \tag{3.7}$$

$$W_\mathrm{B} = 2ur \approx 0.4270509831, \tag{3.8}$$

$$W_\mathrm{M} = r^2 \approx 0.0954915028. \tag{3.9}$$

The three weights are exact once A3 is accepted. Their recurring appearance in later constructions is therefore a real consistency constraint: a later calculation cannot change their ratios without changing the foundation. Recurrence is not, however, independent evidence when the later formula was built from the same partition. The edition will distinguish *reuse of a prior* from *recovery from independent data*.

Table 3.1.. The exact partition and the limits of its interpretation.

Term	Exact role	Physical status
$W_{\mathrm{L}} = u^2$	diagonal outward contribution	internal algebra; "logical" is a label
$W_{\mathrm{B}} = 2ur$	mixed cross term	internal algebra; boundary/interference meaning is conditional
$W_{\mathrm{M}} = r^2$	diagonal retained contribution	internal algebra; matter or mass meaning is conditional

3.4. A minimal dependency graph

The numerical chain is short enough to display without rhetoric:

$$\begin{gathered} \boxed{M} \Longrightarrow \boxed{\varphi^2 = \varphi + 1}, \\ \xrightarrow{\text{A3: two-sided normalization}} \boxed{r = (2\varphi)^{-1},\ u = 1 - r} \Longrightarrow \boxed{u^2, 2ur, r^2}. \end{gathered} \tag{3.10}$$

The only free conceptual step in the displayed numerical calculation is the choice and physical interpretation of M and A3. Everything to the right is exact algebra. That is the strongest scientifically defensible way to state the result.

CHAPTER 4

Traversal, closure, and records

4.1. Traversal as an ordered map

A traversal is not simply an edge between two labelled vertices. It is a composable sequence

$$\tau = (x_0 \xrightarrow{f_1} x_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} x_n) \tag{4.1}$$

together with the order in which its maps act. Two traversals can share endpoints and still differ through their intermediate states, their accumulated phase, their boundary data, or the records they leave. This elementary observation is the origin of the corpus's later insistence that bulk equations alone may be a coarse transcript of a physical action.

Let $\mathcal{O}$ denote the admissible future operations and $R : X \to \mathcal{Y}$ a readout. A distinction between x and x' is persistent over $\mathcal{O}$ if

$$\forall O \in \mathcal{O}_{\text{relevant}}, \qquad R(Ox) \neq R(Ox'). \tag{4.2}$$

The definition is relative to a readout and a future-operation class. In noisy physics one normally replaces the sharp inequality by a statistical discriminability criterion. The binary record of the inherited corpus is therefore an ideal limiting case.

DEFINITION 4.1 — RECORD

A record is a physical state whose distinction from the relevant alternatives remains recoverable under a declared family of subsequent operations and readouts.

This definition accommodates a detector latch, a memory bit, a track in a chamber, a durable macroscopic correlation, or a stored data product. It also reveals why a record cannot be inferred from a label: persistence is a dynamical property.

4.2. Closure and return

The corpus distinguishes *return* from *closure*. A path returns when its projected endpoint agrees with its starting point. It closes as a physical record only if all state variables relevant to identity—including order, phase, boundary data, or memory—satisfy the stated closure criterion. A periodic coordinate can therefore hide a nonperiodic history.

If $\pi : \widetilde{X} \to X$ forgets a history coordinate, then it is possible that

$$\pi(\tilde{x}_n) = \pi(\tilde{x}_0) \qquad \text{while} \qquad \tilde{x}_n \neq \tilde{x}_0. \tag{4.3}$$

This is ordinary mathematics of a lift or covering representation. Volume III asks when such a lifted coordinate is physically measurable and whether it can be made covariant.

4.3. The proposed crossing quantum

The inherited register associates one successful primitive crossing with a retained factor r, and a minimal three-crossing closed history with

$$r^3 = \frac{1}{8\varphi^3} = \frac{\sqrt{5}-2}{8} \approx 0.0295084972. \tag{4.4}$$

The identity is exact. The quantisation claim is not yet a theorem. The binary definition of a record tells us when the chosen readout treats a distinction as recovered; it does not entail that nature assigns the numerical cost r to every successful crossing.

Exposure 4.1 — missing cost functional

To derive Equation (4.4) as physics, the programme must define a cost, probability, amplitude, or conserved current for a primitive crossing; show why its value is exactly r; and demonstrate why three crossings, rather than any other admissible history, govern the target observable. Until then, r^3 is an exact internal coordinate and a candidate parameter-free threshold.

4.4. Retention and the arrow

The corpus's retention principle says that a continued self-description joins a new resolution to the existing record rather than replacing that record. This is a coherent identity criterion: if the state at step $n+1$ contains no recoverable relation to the state at step n, the phrase "the same system continued" has lost its operational basis.

The principle does not prove a universal thermodynamic arrow. Microscopic reversible dynamics, open-system entropy production, cosmological time orientation, and the psychological arrow are different questions. Landauer's bound connects logically irreversible erasure to thermodynamic cost under explicit physical assumptions (Landauer, 1961); Bennett showed how reversible computation clarifies that connection (Bennett, 1982). Those results support careful separation of logical and thermodynamic irreversibility, not their automatic identification.

CONDITIONAL IDENTIFICATION 4.1 — ORDER AS PHYSICAL TIME

If persistent record formation is monotone along a single physical parameter, that parameter supplies an operational time orientation for the recording subsystem. Establishing a unique global time for an arbitrary spacetime requires additional causal and geometric hypotheses.

CHAPTER 5

From evolution to action

5.1. The action principle and its quotient

For fields ϕ^a on a domain M, an action is a functional

$$S[\phi] = \int_M L(x, \phi, \partial\phi, \ldots)\, \mathrm{d}^d x. \tag{5.1}$$

Its Euler–Lagrange expressions are

$$E_a(L) = \sum_{|I|\geq 0} (-D)_I \left(\frac{\partial L}{\partial(D_I \phi^a)} \right). \tag{5.2}$$

Two Lagrangian densities can give the same bulk equations. Locally, adding a total divergence,

$$L' = L + D_\mu B^\mu, \tag{5.3}$$

does not change Equation (5.2) for compactly supported variations. This motivates an equivalent-action relation

$$L \sim_{\mathrm{EA}} L' \quad\Longleftrightarrow\quad E(L) = E(L'). \tag{5.4}$$

Noether's theorem relates differentiable variational symmetries to conserved currents (Noether, 1918). The variational bicomplex gives a systematic language for Euler–Lagrange forms, horizontal divergences, and the inverse problem (Anderson, 1992; Olver, 1993). The important scientific point is negative as well as positive: not every differential equation is Euler–Lagrange in its given variables. The Helmholtz conditions test whether a source form is locally variational.

Established result 5.1 — the inverse-variational gate

A system of differential equations is locally Euler–Lagrange only when its linearisation satisfies the appropriate formal self-adjointness/Helmholtz conditions, possibly after a nondegenerate multiplier is introduced. Dissipative equations such as the heat equation do not generally arise from an ordinary local action of the same field alone.

5.2. Why bulk equivalence can be too coarse

Equation (5.4) forgets boundary data. In general relativity the Einstein–Hilbert action contains second derivatives of the metric; a well-posed Dirichlet variational principle on a manifold with boundary requires the Gibbons–Hawking–York term (York, 1972; Gibbons and Hawking, 1977). Two expressions may therefore agree in their bulk Euler–Lagrange equations yet differ in boundary variation, conserved charge, semiclassical action, or thermodynamic interpretation.

This gives a rigorous core to the corpus's "transcription" critique. A transcript is adequate for a target question only if it preserves every distinction on which the answer depends.

INTERNAL RESULT 5.1 — SUFFICIENCY OF A COMPRESSION

Let $C : X \to Z$ be a compression and $Q : X \to Y$ a target quantity. Then Q is exactly recoverable from C if and only if Q is constant on every fibre of C, equivalently if and only if there exists $\widetilde{Q} : Z \to Y$ such that

$$Q = \widetilde{Q} \circ C. \tag{5.5}$$

This is the precise content of the corpus's field-sufficiency rule.

If two physical preparations collapse to the same compressed description but later produce different target values, the compression is harmful for that target. If no admissible future operation separates them, the collapse is harmless relative to the declared experiment.

5.3. The principle of logical action

The corpus proposes a refinement of bulk action equivalence. In addition to the Euler–Lagrange operator, a complete physical transcript may need to retain

$$\Lambda[L] equiv\big(E(L),\, \Theta_L,\, \mathscr{B}_L,\, \mathscr{O}_L,\, \mathscr{R}_L\big), \tag{5.6}$$

where Θ_L is the symplectic-potential/boundary variation, $\mathscr{B}_L$ denotes boundary conditions and boundary terms, $\mathscr{O}_L$ records execution or causal ordering when relevant, and $\mathscr{R}_L$ denotes the readout relations the action is intended to preserve. The exact list is programme-dependent; Equation (5.6) makes that dependence inspectable.

Definition 5.1 — logical-action equivalence

For a declared target class $\mathcal{Q}$, write $L \sim_{\mathrm{LA}} L'$ when their complete traces agree on every component required to recover all $Q \in \mathcal{Q}$. Logical-action equivalence is therefore finer than equality of bulk equations whenever boundary, order, or record data affect the target.

This definition avoids claiming that one universal maximal transcript has already been found. The scientific task is to state the target first and prove sufficiency by factorisation as in Equation (5.5).

Research target 5.1 — minimal physical transcript

Classify, for major field theories, the smallest extension of the Euler–Lagrange quotient that preserves boundary charges, causal composition, measurement couplings, and empirically distinguishable histories. A successful classification would turn the corpus's Principle of Logical Action from a methodological rule into a representation theorem.

CHAPTER 6

The quantum interface

6.1. What ignorance does not prove

Suppose two alternatives remain compatible with the available record. One may represent that situation by a classical probability distribution, a coherent quantum state, a set of possible models, or a more general convex state. The absence of a distinguishing event does not select among those structures. Accordingly, the inherited statement "superposition is forced" is too strong if based only on nonselection.

A quantum reconstruction needs operational assumptions that distinguish coherent composition from classical mixing. Examples include continuous reversible transformations, local tomography, purification, or interference structure; modern reconstructions make such assumptions explicit (Chiribella et al., 2011). The framework may seek to motivate them through traversal and retention, but it cannot omit them.

EXPOSURE 6.1 — SUPERPOSITION IS NOT A CONSEQUENCE OF IGNORANCE

The proposition "no event selected an alternative" is compatible with both a density matrix containing coherence and a classical mixture diagonal in the same basis. An empirical or operational axiom sensitive to interference is required to distinguish them.

6.2. When complex phase follows

There is nevertheless a clean conditional result behind the corpus's phase argument. Let an additive action parameter $S \in \mathbb{R}$ label sequential composition, and let $\chi : \mathbb{R} \to U(1)$ be a continuous homomorphism:

$$\chi(S_1 + S_2) = \chi(S_1)\chi(S_2), \qquad |\chi(S)| = 1. \tag{6.1}$$

Then

$$\chi(S) = e^{i\kappa S} \tag{6.2}$$

for some real constant κ. With $\kappa = 1/\hbar$, Equation (6.2) is the familiar quantum phase assigned to an action history, central to Feynman's spacetime formulation

(Feynman, 1948).

Internal result 6.1 — continuous phase character

Complex exponential phase is the unique continuous one-dimensional unitary representation of additive real action, up to the scale κ. The premises already contain the unit circle and reversible multiplicative composition; the theorem does not derive those premises or the value of $\hbar$.

6.3. Probability: the precise Gleason route

Born introduced the probabilistic interpretation of quantum amplitudes in scattering theory (Born, 1926). Gleason later proved that, on a real or complex Hilbert space of dimension at least three, a countably additive nonnegative measure on the lattice of closed subspaces has the trace form

$$p(P) = \mathrm{Tr}(\rho P) \tag{6.3}$$

for a positive trace-class operator ρ (Gleason, 1957). Normalization fixes $\mathrm{Tr}\,\rho = 1$. Generalized-measurement versions obtain the trace rule for effects, including qubit settings under the corresponding assumptions (Busch, 2003; Caves et al., 2004).

Established result 6.1 — noncontextual quantum probability

For the measurement domains covered by Gleason-type theorems, additive probabilities assigned noncontextually to the same projector or effect have the Born trace form. The theorem presupposes Hilbert-space event structure and noncontextual additivity; it is not a derivation of Hilbert space from the seven-operation grammar.

The corpus's phrase "the same physical effect carries the same probability in every complete context" is a readable statement of measurement noncontextuality. The final edition retains that intuition but places the theorem's dimension and event-space hypotheses beside it.

6.4. Symmetry, unitarity, and the generator

Wigner's theorem states that a bijection of rays preserving transition probabilities is induced by a unitary or antiunitary transformation (Wigner, 1959). A continuous one-parameter family connected to the identity is represented unitarily. Stone's theorem then states that every strongly continuous one-parameter unitary group $U(t)$ has a unique self-adjoint generator H such that

$$U(t) = e^{-itH} \tag{6.4}$$

in mathematical units (Stone, 1932). Restoring $\hbar$ and identifying H with energy gives

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} |\psi(t)\rangle = H |\psi(t)\rangle . \tag{6.5}$$

ESTABLISHED RESULT 6.2 — THE WIGNER–STONE CHAIN

Transition-probability preservation plus bijectivity gives a unitary or antiunitary implementation; strong continuity of a one-parameter unitary group gives a self-adjoint generator. The physical identification of the parameter with time and the generator with energy is additional structure.

6.5. Measurement and record formation

An ideal premeasurement correlates a system basis with apparatus pointer states,

$$\left(\sum_i c_i |s_i\rangle \right) |M_0\rangle \longrightarrow \sum_i c_i |s_i\rangle |M_i\rangle . \tag{6.6}$$

This unitary interaction does not by itself select one term. Environmental decoherence can suppress interference between robust pointer records in reduced states, explaining the practical stability of classical alternatives (Zurek, 2003); general instruments describe conditioned state updates and outcome statistics (Ozawa, 1984). Neither fact licenses collapsing premeasurement, decoherence, conditioning, and human awareness into one unnamed event.

The framework's observer definition is useful precisely when made mechanical: a prepared ready state, a coupling that defines the channel, propagation, amplification, resolution, and a persistent record. Volume III develops that chain and the information lost at each stage.

6.6. The honest quantum ledger

Table 6.1.. What the foundational chain does and does not establish.

Statement	Grade	Required hypotheses or missing bridge
Continuous additive action has exponential phase	internal conditional result	representation into $U(1)$, continuity, reversible composition
Born trace rule	established external result	Hilbert effects/projectors, additivity, noncontextuality, dimensional conditions
Closed continuous evolution has a self-adjoint generator	established external result	ray symmetry, bijectivity, strong continuity, one-parameter group
The seven-operation grammar uniquely yields quantum theory	open	operational reconstruction including composition, interference, tomography, and state-space axioms
Definite outcomes follow from record persistence	open	interpretive measurement model beyond unitary premeasurement and decoherence

CHAPTER 7

Information, compression, and bounded recovery

7.1. Records are outputs of channels

Let a world state x generate a record y through a measurement channel M:

$$y = M(x) + \epsilon, \tag{7.1}$$

where ϵ represents declared noise or model discrepancy. Any distinction in the exact kernel of a linear M, or more generally in a fibre of a nonlinear M, is absent from the record. No decoder using y alone can determine which member of that fibre occurred. This is a direct consequence of factorisation, and in probabilistic settings it is expressed by data-processing inequalities in information theory (Shannon, 1948; Cover and Thomas, 2006).

ESTABLISHED PRINCIPLE 7.1 — NO RECOVERY WITHOUT TRANSMITTED INFORMATION OR A PRIOR

If two admissible states induce the same distribution of records, no estimator based only on those records can distinguish the states. Recovery can become possible only by changing the channel, adding data, or restricting the admissible state ensemble.

This principle disciplines the corpus's observational language. A telescope, detector, simulation catalogue, and human classification do not reveal "the object itself." Each yields a record through a mechanism with a passband, noise model, point-spread function, selection function, and null space.

7.2. Recovery under a prior

Compressed sensing demonstrates how an underdetermined linear problem can become well posed when the unknown is sparse in a known representation and the measurement operator satisfies suitable conditions (Donoho, 2006;

Candès et al., 2006). Schematically,

$$\hat{x} = \arg\min_{z} \|z\|_1 \quad \text{subject to} \quad \|Mz - y\|_2 \leq \delta. \tag{7.2}$$

The prior does not restore information destroyed for an unrestricted state. It restricts the candidate ensemble so that the surviving measurements identify one member of that smaller class.

Unified Mechanics proposes its channel grammar and fixed weights as such a structural prior. That is a coherent programme, but it is not automatically compressed sensing. For each application the edition will ask:

1. What is the measurement map M?
2. What class of states is restricted by the prior?
3. What theorem makes the inverse unique or stable on that class?
4. How many adjustable parameters enter the map to the observable?
5. Was the map fixed before inspecting the evaluation data?

7.3. The zero-knob rule

A parameter-free map can make a sharp forecast, but "zero fitted parameters" is not sufficient for evidential force. A mapping may contain hidden discrete choices, selected normalisations, flexible data cuts, or an observable chosen after inspection. Accordingly, the governing rule is:

DEFINITION 7.1 — ADMISSIBLE FORECAST

A forecast is admissible as parameter-free only when the source-to-observable map, all discrete choices, normalization conventions, data cuts, loss function, and success criterion were fixed before the evaluation data were inspected. Otherwise the result is a retrospective comparison or a postdiction.

The rule applies equally to positive and negative outcomes. A non-identifiable test should be rejected before execution when possible. A test whose failure is reinterpreted only afterward does not become informative through narrative repair.

7.4. Harmless and harmful quotients

Let $C : X \to Z$ collapse states into equivalence classes. For a family of future quantities $\mathcal{Q}$, the collapse is harmless if every $Q \in \mathcal{Q}$ factors through C. It is harmful if some admissible target separates two states in one fibre:

$$\exists x, x', Q : quad C(x) = C(x') \quad \text{and} \quad Q(x) \neq Q(x'). \tag{7.3}$$

This criterion unifies several themes in the corpus: boundary terms erased by the bulk Euler–Lagrange map, phases erased by an incoherent mixture, traversal depth erased by a periodic projection, and physical distinctions erased by a sensor.

INTERNAL RESULT 7.1 — TARGET-RELATIVE KERNEL STABILITY

A kernel is harmless relative to a future dynamics F_t and readout R when $C(x) = C(x')$ implies $R(F_t x) = R(F_t x')$ for every relevant t. It is harmful when the implication fails. This is a target-relative criterion, not an absolute property of the compression.

CHAPTER 8

The foundational register

8.1. Minimal commitments

The reconstructed foundation can be stated without importing later particle, gravitational, or cosmological claims.

Table 8.1.. Foundational commitments and their living-edition status.

ID	Name	Status and content
A1	systemhood	framework postulate: internal difference is compatible with operational unity
A2	ground	ontological postulate: retained distinction supplies the first operational locus
A3	two-sided reading	numerical postulate fixing the factor of two in $r = (2\varphi)^{-1}$
A4	saturation	open physical postulate requiring a defined capacity and threshold
A5	conserved read-out	physical postulate requiring a measure or current in each realisation
A6	admissible reduction	operator-algebra definition: unital, idempotent, completely positive, bimodular, and covariant

The first three generate the conceptual and numerical foundation. A4 and A5 become relevant only with dynamics. A6 belongs to the carrier and readable-algebra construction of Volume II.

8.2. Core definitions

TABLE 8.2.. Core vocabulary, condensed from the inherited register.

ID	Term	Operational definition
D1	distinction	a difference recoverable by at least one declared test
D2	identity	recoverability of a distinction under admissible change
D3	relation	comparison, coupling, or transformation without automatic identification
D4	seven doings	distinguish, hold, close, answer, change, order, record
D5	traversal stages	persist, distinguish, relate, propagate, resolve, return, close
D6	object	a stable equivalence class under declared transformations and readouts
D7	update	$(a,b) \mapsto (a+b,a)$, the chosen two-channel recurrence
D8	channels	outward share u and retained share r, with $u+r=1$
D9	weights	$W_{\mathrm{L}} = u^2$, $W_{\mathrm{B}} = 2ur$, $W_{\mathrm{M}} = r^2$
D10	crossing	a primitive enacted traversal; its proposed price is not yet derived
D11	record	a physically retained distinction recoverable under relevant future operations
D12	observer	a system that couples to, resolves, amplifies, and persistently retains another distinction
D13	carrier	the mathematical structure on which records are represented; self-duality is a framework requirement
D14	primitive crossing	a smallest nonzero admissible carrier displacement, once a carrier and norm are specified
D15	cluster projection	admissible reduction onto a readable subalgebra
D16	decay generator	a positive operator whose kernel is the readable algebra; minimal candidate $I-C$
D17	readable algebra	the fixed/range algebra retained by the reduction

ID	Term	Operational definition
D18	geometry projectors	proposed mutually orthogonal external, temporal, and closure sectors

8.3. What Volume I establishes

Internal synthesis

The specified update has the golden ratio as its stable positive eigenvalue. With the additional two-sided normalization A3 it fixes r, u, and the three quadratic weights exactly. The framework also supplies a target-relative language of identity, records, compression, and traversal, plus a refined action-equivalence programme that keeps boundary and order data when a target depends on them.

Foundational obligations carried forward

The series has not derived the completeness of the seven-operation grammar, the physical value of a crossing, the uniqueness of a global time mode, Hilbert-space quantum theory from the grammar alone, or a universal minimal logical-action transcript. These are explicit research obligations, not conclusions hidden behind later notation.

8.4. Forward link

Volume II asks whether the abstract carrier requirements—finite primitive crossings, self-duality, an admissible reduction, and a readable algebra—select a concrete exceptional structure. That step introduces lattice integrality, evenness, rank, root systems, Coxeter geometry, and field-theoretic identifications. None of those properties is smuggled into the foundation retroactively; each appears where its new assumption or theorem is first required.

PART II

Carrier Geometry, Fields, and Gravitation

CHAPTER 9

The carrier problem

9.1. From abstract traversal to a concrete representation

Volume I defined a traversal without specifying the set on which it acts. A concrete carrier must add at least four ingredients:

1. a state or displacement space;
2. a composition law for admissible steps;
3. an inner product, norm, or other invariant that distinguishes primitive from composite steps;
4. a symmetry group under which equivalent descriptions agree.

The corpus proposes a finite, self-dual carrier and identifies its primitive crossings with the shortest nonzero vectors of a lattice. That proposal is mathematically productive, but the logical order matters. Self-duality alone does not imply that the carrier is a positive-definite integral lattice; integrality does not imply evenness; and none of these properties fixes the rank. They are separate premises or intermediate obligations.

DEFINITION 1.1 — LATTICE CARRIER

A lattice carrier is a full-rank discrete subgroup $L \subset \mathbb{R}^n$ equipped with the Euclidean inner product. Its dual is

$$L^* = \{x \in \mathbb{R}^n : \langle x, y \rangle \in \mathbb{Z} \text{ for every } y \in L\}. \tag{9.1}$$

It is *unimodular* when $L = L^*$, and *even* when $\langle x, x \rangle \in 2\mathbb{Z}$ for all $x \in L$.

Positive-definite even unimodular lattices exist only in dimensions divisible by eight. In dimension eight there is, up to isometry, one such lattice: E_8 (Conway and Sloane, 1999). This is a strong classification theorem. It becomes a carrier-selection theorem only after the framework derives or stipulates positive definiteness, integrality, evenness, unimodularity, and minimal dimension eight.

Conditional selection 1.1 — the first even unimodular carrier

If the carrier is required to be a positive-definite even unimodular lattice, and if the realised rank is the smallest positive admissible rank, then it is isometric to the E_8 lattice. The mathematical uniqueness is established; the physical premises remain framework commitments.

9.2. Why the inherited evenness argument is incomplete

The corpus motivates even norm by saying that a relation read outward and back contributes twice to its self-pairing. This can become a proof only after the pairing is defined and shown to take integer values on all carrier vectors. Without integrality, "twice" gives a real number, not an even integer. Without a construction showing that every lattice vector is assembled from such two-sided relations, the argument covers only selected generators.

Accordingly, this edition treats evenness as an open representation lemma:

Research target 1.1 — derive the lattice hypotheses

Construct a functor or representation from the traversal grammar to a lattice L and prove, rather than name, that (i) the pairing is integral, (ii) all norms are even, (iii) $L = L^*$, (iv) the form is positive definite, and (v) the minimal faithful rank is eight. Only then would the E_8 selection be derived from the foundation.

CHAPTER 10

The E_8 root carrier

10.1. Coordinates and primitive crossings

In a standard normalization the E_8 roots have squared norm two. They consist of two families:

$$\Phi_{mint} = \{(\pm 1, \pm 1, 0, \dots, 0) \,:\, \text{all permutations}\}, \tag{10.1}$$

$$\Phi_{\text{half}} = \left\{\frac{1}{2}(\pm 1, \dots, \pm 1) \,:\, \text{an even number of minus signs}\right\}. \tag{10.2}$$

There are 112 roots in the first family and 128 in the second, for a total of 240. Their integral span is the E_8 root lattice. The root system is invariant under the Weyl group generated by reflections

$$s_\alpha(x) = x - 2\frac{\langle x, \alpha\rangle}{\langle \alpha, \alpha\rangle}\alpha. \tag{10.3}$$

The corpus identifies these roots with primitive crossings. Mathematically, they are the shortest nonzero lattice vectors. Physically, the identification requires an action or transition rule showing why a shortest lattice vector is an indivisible event rather than merely a convenient coordinate step.

10.2. Triangular closure

If roots $\alpha, \beta \in \Phi(E_8)$ satisfy $\langle \alpha, \beta\rangle = -1$, then $\alpha + \beta$ is also a root. Setting $\gamma = -(\alpha + \beta)$ gives

$$\alpha + \beta + \gamma = 0. \tag{10.4}$$

The parallelogram spanned by α and β has Gram determinant

$$\det\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3, \tag{10.5}$$

so its area is $\sqrt{3}$, while the triangle bounded by the three roots has area $\sqrt{3}/2$. Earlier corpus language sometimes called both the root cell. The distinction is fixed here.

Internal result 2.1 — minimal nondegenerate root polygon

Within an admissible-step model in which roots are steps and immediate reversal is excluded as degenerate, a three-root relation $\alpha + \beta + \gamma = 0$ is the shortest possible closed polygon. Its geometric existence is exact. Its interpretation as the minimal history of physical closure is conditional on the step model.

If one attaches the proposed retained factor r to each step, the three-step product is the exact number $r^3 = (\sqrt{5} - 2)/8$. This is reuse of the foundational prior, not an independent recovery.

10.3. Complete finite closures

Because every root occurs with its negative,

$$\sum_{\alpha \in \Phi(E_8)} \alpha = 0. \tag{10.6}$$

Thus a sequence using every root exactly once has zero net displacement. The corpus also supplies a computational witness satisfying its additional local sequencing rule. Volume V records the witness, code, hash, and verifier. Existence of one such ordering does not establish uniqueness, count all admissible closures, or show that nature executes one.

10.4. The Coxeter plane and golden structure

The E_8 Coxeter element has order thirty. Projection of the root system onto an associated Coxeter plane places the 240 roots on eight rings of thirty. In the corpus's numerical certificate, the ring radii pair into four ratios equal to φ within floating-point tolerance. This geometry belongs to established relationships among Coxeter systems, quasicrystalline projections, and the noncrystallographic group H_4 (Elser and Sloane, 1987; Koca et al., 1998; Coxeter, 1973).

Established contact 2.1 — E_8, H_4, and the golden field

The E_8 lattice admits four-dimensional cut-and-project descriptions with H_4 symmetry, and quaternionic formulations decompose its roots into golden-related H_4 sets (Elser and Sloane, 1987; Koca et al., 1998). The golden ratio is therefore genuine carrier geometry; its presence does not establish a physical vacuum or coupling.

CHAPTER 11

Finite spherical designs

11.1. The 24-cell and 600-cell

The 24-cell is the regular four-polytope associated with the D_4 roots, and the 600-cell has 120 vertices forming the H_4 root system. A finite set $X \subset S^{d-1}$ is a spherical t-design if

$$\frac{1}{|X|} \sum_{x \in X} p(x) = \frac{1}{\operatorname{vol} S^{d-1}} \int_{S^{d-1}} p(x) \, \mathrm{d}\Omega \tag{11.1}$$

for every polynomial p of total degree at most t (Delsarte et al., 1977). Exact coordinate checks in the corpus give

$$X_{24} \text{ is a spherical 5-design,} \qquad X_{600} \text{ is a spherical 11-design.} \tag{11.2}$$

If harmonics of degree N are sampled on such a set, their Gram products have degree at most $2N$. Exact orthogonality therefore holds when $2N \leq t$. For Fock's hydrogenic momentum-space harmonics, $N = n - 1$, so the sampling limits are

$$t = 5 : \quad n \leq 3, \tag{11.3}$$

$$t = 11 : \quad n \leq 6. \tag{11.4}$$

ESTABLISHED RESULT 3.1 — WHAT THE CUTOFFS MEAN

The observed finite sampling cutoffs follow from spherical-design strength. This explains why the discrete Gram matrix is exact through those degrees and fails beyond them. It is a cubature theorem, not a derivation of hydrogen or of the Coulomb interaction.

The physical application and its circularity are treated in Volume IV. Here the important result is methodological: the loss of exactness at $n = 4$ or $n = 7$ is the predictable resolution limit of the corresponding finite design.

CHAPTER 12

Admissible reduction and the readable algebra

12.1. The operator-algebraic construction

Let $\mathscr{A}$ be a unital C^*-algebra of carrier observables and $\mathscr{B} \subseteq \mathscr{A}$ a unital subalgebra. The corpus calls a reduction $C : \mathscr{A} \to \mathscr{B}$ admissible when it is unital, idempotent, completely positive, bimodular over $\mathscr{B}$, and covariant under the declared carrier symmetries. Define the minimal decay generator

$$K_0 = I - C. \tag{12.1}$$

Then

$$\ker K_0 = \operatorname{ran} C = \mathscr{B}. \tag{12.2}$$

Equation (12.2) is exact once C is specified. Calling $\mathscr{B}$ the physical readable algebra is the framework's interpretation.

A norm-one projection from a C^*-algebra onto a C^*-subalgebra has the bimodule property of a conditional expectation, under the hypotheses of Tomiyama's theorem (Tomiyama, 1957). For a finite group G acting by automorphisms, the canonical fixed-point expectation is

$$C_G(a) = \frac{1}{|G|} \sum_{g \in G} g \cdot a, \tag{12.3}$$

whose image is $\mathscr{A}^G$.

Established result 4.1 — finite-group averaging

Equation (12.3) is unital, positive, idempotent, covariant, and maps onto the fixed algebra. Uniqueness requires the action, target algebra, and covariance/conditional-expectation class to be fixed; covariance in the abstract does not select a group by itself.

12.2. Minimal and graded decay

Since $C^2 = C$, the minimal generator $K_0 = I - C$ is also idempotent. Its spectrum lies in $\{0, 1\}$. That is adequate for separating readable from suppressed sectors, but too coarse for a nontrivial hierarchy of masses or decay rates. A general positive generator K may satisfy

$$K\mathbf{1} = 0, \qquad \ker K = \mathscr{B}, \tag{12.4}$$

while carrying more than two eigenvalues. The corpus correctly distinguishes these tasks. The later closure analysis locates the mass-squared operator in the boundary 15 and reduces numerical family splitting to a twelve-component stationary selector. Determining that stationary spectrum, together with its absolute scale, remains the localized calculation beyond the minimal projector K_0.

12.3. The rank-three candidates

Up to the relevant Weyl conjugacy, the inherited finite enumeration identifies three rank-three root-subsystem types and their rank-five orthogonal complements:

TABLE 12.1.. Rank-three candidates in the corpus enumeration.

Decayed subsystem	Orthogonal complement	Root count of complement
$A_1 \oplus A_1 \oplus A_1$	$D_4 \oplus A_1$	26
$A_2 \oplus A_1$	A_5	30
A_3	D_5	40

The corpus selects A_3: the first candidate lacks an A_2-type triangular root history, while the second is rejected as reducible. The first exclusion is meaningful within the adopted history rule. The second requires more care: a reducible root system does not, by itself, prove that a physical reduction must split into two independent generators. That is a model assumption about how algebraic factors map to dynamics.

CONDITIONAL SELECTION 4.1 — A_3 DECAY AND D_5 SURVIVOR

If the physical reduction is a single irreducible reflection-generated decay, if admissible decay must contain the selected triangular history, and if the rank bookkeeping of the geometry block is accepted, the surviving root algebra is $D_5 \cong \mathfrak{so}(10)$. The root-complement calculation is exact; the physical selection premises remain conditional.

12.4. The $A_3 \times D_5$ branching

At the level of complex representations, the E_8 adjoint admits the dimensionally consistent branching

$$\mathbf{248} = (\mathbf{15}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{45}) \oplus (\mathbf{6}, \mathbf{10}) \oplus (\mathbf{4}, \mathbf{16}) \oplus (\overline{\mathbf{4}}, \overline{\mathbf{16}}). \tag{12.5}$$

The **16** of $SO(10)$ is the standard representation accommodating one Standard Model family together with a right-handed neutrino (Fritzsch and Minkowski, 1975). Equation (12.5) explains why matter-like spinor sectors appear in the finite carrier bookkeeping.

Exposure 4.1 — representation content is not yet matter

A root decomposition does not supply a Lorentz-covariant quantum field, spin-statistics, chirality, a kinetic term, a gauge coupling, or a mass matrix. The claimed split of a **4** index into one invariant direction plus three families is a proposed physical reading, not a standard consequence of Equation (12.5).

Direct attempts to embed Lorentz and Standard Model gauge structure in a real or complex form of E_8 face representation-theoretic no-go results. Distler and Garibaldi state precise subgroup and chirality hypotheses for the class they exclude (Distler and Garibaldi, 2010). The corpus argues that its finite decay group is not the Lorentz subgroup assumed there. That observation is not yet an evasion: a complete model must first provide its own Lorentz action and chiral field content, after which the theorem's applicability can be assessed exactly.

CHAPTER 13

Connections, holonomy, and the co-distinction proposal

13.1. Established geometric structure

On a principal bundle, a connection one-form A specifies parallel transport and has curvature

$$F = \mathrm{d}A + A \wedge A. \tag{13.1}$$

For a small loop $\gamma = \partial\Sigma$, holonomy differs from the identity at leading order by the curvature flux through Σ. In a gauge theory, A and F take values in a Lie algebra; Yang and Mills introduced the non-Abelian field strength and its gauge dynamics in physics (Yang and Mills, 1954). Geometric phase supplies another physical manifestation of holonomy (Berry, 1984; Aharonov and Anandan, 1987).

The corpus's useful conceptual statement is that ordered transport can fail to close even when the projected endpoint returns. Equation (13.1) already formalizes this for connections. The new proposal is to treat distinguishable actions themselves as the transported quantity and to select among their cycles by a least-cost rule.

13.2. The Shields field as a proposed extension

Let a denote a point in a parameter or action space, and let $\mathscr{A}^{\alpha}{}_{i}(a)$ be a connection-like object mapping changes in that space to a co-distinction displacement. The proposed loop response is

$$\Delta X^{\alpha}[\gamma] = \oint_{\gamma} \mathscr{A}^{\alpha}{}_{i}(a)\,\mathrm{d}a^{i}. \tag{13.2}$$

Nonzero loop response is called a radiance in the corpus; variation across paths is associated with scattering or curvature. These names are not yet physical fields. A field theory requires a base manifold, fibre, transformation law, curvature, source, action, dimensions, and coupling to instruments.

Research target 5.1 — least-cost holonomy

Define a gauge-invariant functional on the holonomy classes of Equation (13.2), prove existence and stability of its minimisers, and derive an observable coupling that distinguishes the proposal from ordinary geometric phase or radiative transport.

Interrogation can change the holonomy class when the measurement coupling becomes part of the loop. That observation is sound but general: a measurement intervention changes the dynamics unless it commutes with them. The new content must lie in a quantitative law for the change.

The gravitational square

14.1. MacDowell–Mansouri gravity

MacDowell and Mansouri formulated gravity using a de Sitter-type gauge connection and a projection to the Lorentz subalgebra (MacDowell and Mansouri, 1977). In schematic first-order notation, decompose the curvature as

$$F^{ab} = R^{ab} - \ell^{-2} e^a \wedge e^b, \qquad F^{a4} = \ell^{-1} T^a, \tag{14.1}$$

where e^a is a coframe, R^{ab} the Lorentz curvature, T^a torsion, and ℓ a length scale. The projected quadratic action contains

$$\int \epsilon_{abcd} \left(R^{ab} \wedge R^{cd} - 2\ell^{-2} R^{ab} \wedge e^c \wedge e^d + \ell^{-4} e^a \wedge e^b \wedge e^c \wedge e^d \right). \tag{14.2}$$

The first term is topological in four dimensions under the usual conditions, the mixed term gives Einstein–Cartan/Einstein–Hilbert dynamics, and the final term is cosmological.

14.2. The corpus coefficient proposal

The corpus forms an abstract defect from a curvature-like object and a realised-area object,

$$\mathscr{D} = u\mathscr{F} - r\mathscr{E}, \qquad \langle \mathscr{D}, \mathscr{D} \rangle = u^2 \langle \mathscr{F}, \mathscr{F} \rangle - 2ur \langle \mathscr{F}, \mathscr{E} \rangle + r^2 \langle \mathscr{E}, \mathscr{E} \rangle. \tag{14.3}$$

The coefficient pattern matches the three-term structure of Equation (14.2). Moreover,

$$\frac{r}{u} = \frac{1}{\sqrt{5}}, \qquad 3\frac{r}{u} = \frac{3}{\sqrt{5}}. \tag{14.4}$$

These are exact internal ratios.

CONDITIONAL IDENTIFICATION 6.1 — FROM THE CARRIER SQUARE TO GRAVITY

Equation (14.3) becomes a MacDowell–Mansouri-type gravitational action only if $\mathscr{F}$, $\mathscr{E}$, the invariant pairing, form degrees, gauge group, symmetry reduction, dimensional scale ℓ, and boundary terms are specified so that the equality holds as an action, not merely as a three-term polynomial pattern.

The corpus further proposes that its cluster projection supplies the symmetry-reducing projection in MacDowell–Mansouri gravity. This is a valuable concrete bridge to test: both maps must be written in a common representation and shown equal or naturally equivalent. At present the identification is proposed, not proved.

14.3. Variation and torsion

For the appropriate first-order gravitational action, independent variation with respect to coframe and connection yields an Einstein equation with cosmological term and a torsion equation. In vacuum or for matter without intrinsic spin, the torsion equation commonly gives $T^a = 0$; spinful matter can source torsion in Einstein–Cartan theory (Hehl et al., 1976). The inherited blanket statement "torsion vanishes rather than being imposed" is therefore too broad.

EXPOSURE 6.1 — FIELD EQUATIONS ARE ONLY AS DERIVED AS THE ACTION MAP

Standard variation of a standard first-order action is borrowed mathematics. It does not validate the corpus action until the map in Conditional Identification 6.1 is complete. Couplings to the proposed matter sector must also be included before the torsion and stress-energy equations can be stated.

CHAPTER 15

Least boundary and the compression field

15.1. The single-cell theorem

Let $\Omega \subset \mathbb{R}^3$ be a bounded region of fixed volume V, let $\eta > 0$ be an isotropic boundary distinction density, and define the instantaneous informational boundary cost

$$\dot{\mathcal{I}}_\partial[\Omega] = W_\mathrm{B}\eta\,|\partial\Omega|. \tag{15.1}$$

The Euclidean isoperimetric inequality gives

$$|\partial\Omega| \geq (36\pi)^{1/3}V^{2/3}, \tag{15.2}$$

with equality for a ball under the regularity hypotheses (Osserman, 1978). Multiplication by the positive constant $W_\mathrm{B}\eta$ preserves the minimiser.

INTERNAL RESULT 7.1 — FINITE-READOUT CELL THEOREM

Given Equation (15.1), Euclidean geometry, isotropic η, and fixed volume, a ball uniquely minimises the boundary cost. If a readout capacity C imposes $W_\mathrm{B}\eta|\partial\Omega| \leq C$, the saturated ball has

$$R = \sqrt{\frac{C}{4\pi W_\mathrm{B}\eta}}. \tag{15.3}$$

The geometry is proved; interpreting η, C, and W_B as physical information variables is conditional.

For a prescribed volume history $V(t)$, applying Equation (15.2) at each time and integrating proves that the pointwise least-boundary history minimizes

$$\mathcal{I}_\partial = \int W_\mathrm{B}\eta\,|\partial\Omega(t)|\,\mathrm{d}t. \tag{15.4}$$

This is a valid conditional variational theorem. It does not derive a physical Lagrangian unless a physical action is shown to represent the same conversion count.

15.2. From isolated cells to saturation

Equal spheres cannot tile Euclidean three-space; their densest packing fraction is $\pi/\sqrt{18}$, as established by the Kepler-conjecture proof (Hales, 2005). A space-filling equal-volume partition is therefore a different variational problem from the isolated isoperimetric problem. Shared interfaces, Plateau conditions, topology, and periodicity enter.

Weaire and Phelan found a partition with lower interface area than Kelvin's candidate (Weaire and Phelan, 1994). It remains a best-known structure, not a proved global optimum. The inherited text is correct to flag this but too strong when it says the saturated shape is proved.

Conditional identification 7.1 — saturated least-interface field

If persistent resolution requires a gap-free equal-volume partition and if total shared interface is the complete cost, saturation selects a minimiser of the three-dimensional Kelvin problem. The minimiser's existence class and local Plateau laws are standard; its global identity is not known. Weaire–Phelan is a candidate, not a theorem.

For anisotropic boundary density $\eta(\hat{n})$, the minimiser is generally a Wulff shape rather than a sphere. This provides a natural falsification boundary: anisotropic fields should deform the cells in a calculable direction.

15.3. The Bubble Field as a compression map

The later Bubble Field is best understood as a finite-dimensional compression, not a new fundamental field by declaration. Its local state is

$$x = (\alpha_{\mathrm{M}}, \alpha_{\mathrm{L}}, \nu, \theta), \tag{15.5}$$

and the compressed variables are

$$C(x) = (A, B, \Theta) = (-\alpha_{\mathrm{M}},\ \alpha_{\mathrm{L}} + \nu/3,\ \theta). \tag{15.6}$$

The Jacobian has rank three and kernel

$$\ker DC = \operatorname{span}\{(0, -1, 3, 0)\}. \tag{15.7}$$

Thus the metric sector cannot separately recover α_{L} and ν. Any claimed output depending on that split needs an additional measurement or constitutive section.

Internal result 7.2 — exact field sufficiency

A quantity $Q(\alpha_\mathrm{M}, \alpha_\mathrm{L}, \nu, \theta)$ is recoverable from the Bubble Field exactly when it is invariant along Equation (15.7), equivalently when $Q = \widetilde{Q}(A, B, \Theta)$. Metric, curvature, geodesic, clock, and lensing formulas written only in A and B satisfy this criterion by construction.

The proposed least-response constitutive section minimizes a Euclidean quadratic cost under the mixed constraint $r\alpha_\mathrm{L} + u\alpha_\mathrm{M} = \chi$, giving the ratio $\alpha_\mathrm{L}/\alpha_\mathrm{M} = r/u = 1/\sqrt{5}$. This is an exact Lagrange-multiplier result for the chosen cost; the cost itself is a testable constitutive assumption.

15.4. Spherical metric reconstruction

In isotropic coordinates the field uses

$$\mathrm{d}s^2 = -e^{2A(\rho)}c^2\mathrm{d}t^2 + e^{2B(\rho)}\left(\mathrm{d}\rho^2 + \rho^2\mathrm{d}\Omega^2\right). \tag{15.8}$$

The exterior Schwarzschild solution is represented exactly by

$$A = \ln\frac{1-x}{1+x}, \qquad B = 2\ln(1+x), \qquad x = \frac{GM}{2\rho c^2}. \tag{15.9}$$

This is a correct coordinate re-expression of established general relativity, originating with Einstein's field theory (Einstein, 1916). It demonstrates that the compression variables can encode a known spherical metric. It does not derive that metric from the microscopic cell variables.

Exposure 7.1 — reconstruction versus dynamics

Equations (15.8)–(15.9) are top-down reconstructions. A bottom-up field theory still requires equations for $\alpha_\mathrm{M}, \alpha_\mathrm{L}, \nu, \theta$, a source action, boundary conditions, and a derivation of the constitutive section.

CHAPTER 16

The source sector and present frontier

16.1. Local and bilocal fields

A local scalar action such as

$$S_\phi = \int \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right] \mathrm{d}^4 x \tag{16.1}$$

has a local stress-energy tensor obtained by metric variation. A bilocal object $\Phi(x, y)$ does not automatically define a local source at x. One must specify a kernel, diagonal restriction, integration over y, or localization map and show covariant conservation of the resulting tensor.

The corpus's source-sector papers correctly isolate this issue. Orbit data determine $\mu = GM$, not G and M separately. Likewise, a geometric reconstruction can identify a metric potential without determining the microscopic source that produced it.

ESTABLISHED IDENTIFIABILITY FACT 8.1

In Newtonian two-body orbital dynamics, trajectories constrain the gravitational parameter $\mu = GM$. Separating G and M requires an independent mass or coupling measurement. The same product degeneracy must be respected by any proposed source reconstruction.

16.2. What is exact, what is proposed

TABLE 16.1.. Carrier and field ledger.

Statement	Grade	Boundary of the result
Uniqueness of rank-eight positive-definite even unimodular lattice	established	conditional carrier selection because the hypotheses are not yet derived
240-root coordinates and root reflections	established	physical step interpretation is additional

Statement	Grade	Boundary of the result
Triangular root closure and areas	exact	physical history/cost is conditional
Golden Coxeter/H4 geometry	established plus certified	no physical coupling follows from geometry alone
Spherical-design sampling limits	exact	cubature, not atomic dynamics
Finite-group fixed-algebra average	established	group and physical algebra must be specified
A_3 complement D_5	exact finite algebra once selected	selection and family reading are conditional
Co-distinction holonomy field	conjecture	requires bundle, action, coupling, and measurement
Three-term gravitational square	exact polynomial pattern	equality to a gravitational action is conditional
Single-cell least-boundary shape	internal theorem from standard geometry	physical information cost is a postulate
Saturated foam identity	open	Weaire–Phelan is best-known, not proven optimal
Bubble compression rank and kernel	exact	dynamics and constitutive section remain conditional

16.3. Forward obligations

The carrier programme has four load-bearing tasks:

1. derive the lattice hypotheses and rank from the traversal grammar;
2. construct the physical reduction and show why it is the A_3 average;
3. write a dimensionally complete action linking the carrier square to gauge, matter, and gravitational fields;
4. derive an observable source and coupling rather than reconstructing known metrics from chosen potentials.

Volume III now turns from the carrier to what can be observed from it: causal order, traversal depth, field assembly, measurement architecture, and the losses introduced by each readout.

PART III

Traversal, Time, Observation, and Readout

CHAPTER 17

Traversal depth

17.1. Recurrence is not return of the complete state

Many problems in the corpus share one mathematical pattern: a projection makes an evolution look closed while a variable omitted by the projection continues to change. Let

$$\pi : \widetilde{X} \to X \tag{17.1}$$

forget a history coordinate. A sequence may satisfy $x_{n+p} = x_n$ in X while its lift satisfies

$$\tilde{x}_{n+p} = T\tilde{x}_n, \qquad T \neq I. \tag{17.2}$$

The projected orbit is periodic; the full orbit carries monodromy. A circle is the projection of a helix when one coordinate is suppressed.

The corpus calls the omitted coordinate *traversal depth*. The name is useful if it is treated operationally: depth is whatever state variable distinguishes repeated projected positions that have different prior histories or future responses. Merely appending an integer counter always removes projected recurrence mathematically. The physical question is whether the counter is covariant, dynamically coupled, and measurable.

INTERNAL RESULT 1.1 — LIFTED NONCLOSURE

Let $h : \widetilde{X} \to \mathbb{R}$ satisfy $h(T\tilde{x}) = h(\tilde{x}) + \Delta h$ with $\Delta h > 0$. Then no orbit with nonzero winding under T is closed in $\widetilde{X}$, even if its projection is closed in X. This is exact mathematics of the chosen lift; it does not establish that nature carries h.

17.2. The helix as a normal form

For a projected angular variable $\theta \sim \theta + 2\pi$, a simple lift is

$$\gamma(s) = \big(R\cos\omega s,\ R\sin\omega s,\ vs\big). \tag{17.3}$$

After one projected period $2\pi/\omega$, the first two coordinates recur and the third advances by $2\pi v/\omega$. The helix is therefore a visualization of Equation (17.2), not a claim that every worldline is literally a corkscrew in an additional spatial dimension.

The helical-relativity documents extend this picture to clock order, holographic scale, phase, and record history. The scientifically safe unification is:

> A recurrent base description may require a nonrecurrent lifted variable whenever the omitted state changes the result of a later admissible operation.

That statement is just the harmful-compression criterion of Volume I applied to recurrence.

17.3. Topology does not supply an arrow

A universal cover distinguishes windings, but topology alone does not privilege increasing over decreasing winding number. Orientation and monotonicity require dynamics, boundary conditions, or a record criterion. Consequently, the corpus's arrow cannot be obtained by drawing a helix. The helix represents an arrow after a monotone history variable has been specified.

EXPOSURE 1.1 — THE MISSING PHYSICAL LIFT

The traversal programme still owes a covariant definition of depth, an evolution equation for it, a coupling to ordinary fields, and an observable that changes when the depth changes. Without those elements, the lift is a faithful bookkeeping extension but not an additional physical degree of freedom.

CHAPTER 18

Causal order and closed timelike curves

18.1. What causal reconstruction establishes

In Minkowski spacetime of dimension greater than two, suitable bijections preserving causal relations are generated by Lorentz transformations together with translations and dilations under the hypotheses of Alexandrov–Zeeman-type results (Zeeman, 1964; Alexandrov, 1967). For sufficiently well-behaved relativistic spacetimes, the family of timelike curves determines conformal and topological structure under distinguishing conditions, as in Malament's theorem (Malament, 1977).

Established result 2.1 — causal structure is geometrically powerful

Causal order can determine the conformal metric structure of a spacetime under specific dimensional, regularity, bijectivity, and causal-distinguishing hypotheses. It does not determine an absolute conformal scale, and the theorem cannot be applied when its causal hypotheses fail.

This reverses a common intuition: geometry is not always prior to causal order. Nevertheless, it does not show that every metric must be derived from the corpus's record order. It shows that a precisely defined causal relation carries substantial geometric information.

18.2. Gödel spacetime as a boundary case

Gödel's rotating cosmological solution of Einstein's equations contains closed timelike curves (Gödel, 1949). It therefore fails standard chronology and distinguishing conditions needed by causal reconstruction. This makes it a useful stress test for any claim that an observer's ordered record is fundamental.

The corpus proposes that a closed timelike curve in the base spacetime lifts to an open curve in record space. The proposal is consistent as an augmentation: the traveller returns to a spacetime event while a history coordinate has advanced. It is not a theorem of general relativity, because general relativity's state is whatever the model includes. Adding the coordinate changes the model.

18.3. Consistency, memory, and identity

Deutsch formulated quantum consistency conditions near closed timelike curves using fixed-point density operators (Deutsch, 1991). Gavassino analysed a closed system on a closed timelike curve and found that global consistency can require recurrence of the complete internal state, including erasure of memories accumulated around the loop (Gavassino, 2024).

The corpus responds that a state whose new record is erased is not a continuation of the same record-bearing identity. This is a legitimate semantic criterion, but it does not invalidate the consistency solution. It changes what the word "same observer" is required to preserve.

CONDITIONAL INTERPRETATION 2.1 — IDENTITY ACROSS A CLOSED TIMELIKE CURVE

If observer identity is defined by persistent recoverability of accumulated records, then a globally consistent loop that erases those records does not return the same record-bearing observer in that sense. This is an identity criterion, not a proof that the spacetime solution is dynamically forbidden.

The inherited "no recollapse while the record stands" claim is likewise not an established result of gravitation. General relativity admits expanding and recollapsing solutions. A no-overwrite law would have to appear as a covariant constraint or stress-energy contribution and survive cosmological tests.

CHAPTER 19

Holographic and scale lifts

19.1. Radial evolution and renormalization-group flow

The AdS/CFT correspondence relates a gravitational theory in an asymptotically anti-de Sitter bulk to a conformal field theory on its boundary in its best-understood settings (Maldacena, 1998; Witten, 1998). The holographic radial coordinate is connected to scale, and Hamilton–Jacobi equations for the bulk action can be written in renormalization-group form (de Boer et al., 2000).

The corpus observes that some conventions express boundary-to-bulk or decreasing-scale evolution with a minus sign. Replacing that sign by a positive "completed history" variable is a reparameterization when the map is monotone. A geometric lift carries more information only if two states at the same projected scale can have distinct histories that later matter.

Let s be a flow parameter and $W(s)$ a monotone functional with

$$\frac{\mathrm{d}W}{\mathrm{d}s} \leq 0. \tag{19.1}$$

Define accumulated evolution

$$E(s) = W(0) - W(s) \geq 0. \tag{19.2}$$

Equation (19.2) is an exact change of variables. It does not add a dimension unless E is retained independently of W and affects observables.

19.2. Finite domains and hidden history

In a finite or periodic base domain, a scale or radial coordinate can recur or saturate while the sequence of coarse-graining operations remains different. A lifted state

$$\tilde{x} = (x, E) \tag{19.3}$$

retains that sequence. This is useful computationally for non-Markovian processes: the enlarged state can make an apparently history-dependent evolution Markovian. Again, the enlargement is justified only when it improves prediction of a declared target.

Research target 3.1 — physical holographic depth

Identify a holographic observable or response function that depends on accumulated traversal history at fixed conventional radial data, and derive the history rate from the bulk/boundary dynamics rather than choosing it. Absent such an observable, the lift is explanatory notation.

CHAPTER 20

Field assembly and lossless transcription

20.1. Assembly is not a list of parts

The corpus uses deliberately artificial examples—including the "Crowley" and "Glorpnorp" constructions—to demonstrate a general point: a collection of component descriptions does not determine the assembled system unless incidence, order, orientation, boundary conditions, and coupling rules are also transmitted.

Formally, let components be $c_1, \ldots, c_n$ and let R denote assembly relations. A source system is

$$X = (\{c_i\}, R). \tag{20.1}$$

A transcript retaining only $\{c_i\}$ identifies all systems with the same parts, regardless of R. If the evolution operator F depends on R, no decoder can reconstruct $F(X)$ from the part list alone.

Internal result 4.1 — non-identifiability from a part list

Let $C(\{c_i\}, R) = \{c_i\}$. If there exist $R \neq R'$ and a relevant evolution/readout Q for which

$$Q(\{c_i\}, R) \neq Q(\{c_i\}, R'), \tag{20.2}$$

then Q does not factor through C. No assembly procedure using only the component list can be lossless for Q.

This theorem is elementary but widely applicable. A field catalogue without spatial adjacency cannot reproduce a gradient-dependent dynamics. A Lagrangian's bulk equation without boundary data cannot reproduce a boundary charge. A periodic coordinate without winding cannot reproduce history-dependent response.

20.2. The evolution-preserving transcription criterion

Let $T : X \to Z$ be a transcript and F_X, F_Z the source and transcript evolutions. The transcript is dynamically sufficient when the diagram commutes:

$$T \circ F_X = F_Z \circ T. \tag{20.3}$$

If only a readout R must be preserved, the weaker target-relative requirement is

$$R_X \circ F_X = R_Z \circ F_Z \circ T. \tag{20.4}$$

The first criterion supports autonomous simulation in the transcript space; the second supports a specific observable.

The corpus's Principle of Logical Action is most rigorous when read as Equation (20.3)*: an action representation is admissible for a task only when its compression commutes with the relevant evolution and readout. It is not necessary to claim that every physical law is literally linguistic.*

20.3. Connection to gauge transport

Comparing fields at separated points requires a rule of transport. A gauge connection supplies precisely that rule; curvature records path-dependent nonclosure. Thus "relation must propagate" has a standard mathematical realization in gauge geometry. The corpus's new claim is not the existence of connections, but the attempt to derive their necessity from lossless comparison. Completing that derivation requires specifying the local equivalence group and proving that any admissible comparison rule is a connection on the corresponding bundle.

CHAPTER 21

Mechanical observation

21.1. The complete measurement chain

A physical description becomes an empirical statement only through a typed chain:

$$\begin{aligned} &\text{mechanism} \to \text{state variable} \to \text{observable} \to \text{coupling} \\ &\quad \to \text{record} \to \text{calibration} \to \text{inference.} \end{aligned} \tag{21.1}$$

Every arrow can discard information or introduce model dependence. The Universopedia layer of the corpus is strongest when used as an audit of Equation (21.1), not as a new force law.

Let X_S be the world-state space at scale S, $D_{S,c}$ the record space of channel c, and

$$M_{S,c} : X_S \to D_{S,c} \tag{21.2}$$

the complete measurement map, including propagation, coupling, selection, instrument response, and noise. A decoder assigns a record to a validated language class, while an inference model estimates physical parameters. These maps must not be conflated.

21.2. Five ledger types

The difference between T and F is crucial. A feature in the transmission kernel cannot be recovered from the same record by a cleverer classifier. A feature in the familiarity gap is present but not understood; it may become classifiable without new data.

21.3. Actual boundaries are channel-dependent

The "actual Boundary" is defined as the causal cut separating retained identity from exchanged information at a selected scale, channel, and timescale. It is not necessarily a material surface. A star has different optical, neutrino, acoustic, gravitational, nuclear, and thermodynamic boundaries. A black-hole event horizon, a photosphere, and a detector threshold are different cuts.

TABLE 21.1.. The observational ledger.

Code	Type	Meaning
R	calibrated record	stored count, image, spectrum, time series, strain, track, or catalogue value
E	established inference	model-dependent physical estimate with units, uncertainty, and validity range
U	framework reconstruction	Unified Mechanics interpretation layered on R and E
T	transmission limit	distinctions the chosen channel did not preserve
F	familiarity gap	structure present in the record but not yet assigned to a validated class

The Sun illustrates the point without requiring a new theory. Photons generated in the interior undergo radiative transport and repeated interaction before escape, whereas neutrinos weakly couple and provide a much more direct channel from nuclear reactions in the core. Solar-neutrino experiments have measured both flux and flavour transformation (SNO Collaboration, 2002; Borexino Collaboration, 2018). The same source can therefore be opaque to one carrier and comparatively transparent to another. Opacity is a property of matter, carrier, energy, and channel together.

21.4. Observer architecture

A mechanical observer is represented by six stages:

1. prepared ready state;
2. channel-defining coupling;
3. propagation through the apparatus;
4. physical resolution or correlation;
5. amplification into robust degrees of freedom;
6. production and retention of a calibrated record.

Human semantic interpretation may follow, but it is not required for physical observation. This preserves the corpus's broad observer definition while keeping measurement dynamics separate from consciousness.

CHAPTER 22

Time, ensembles, and complexity

22.1. Four distinct clocks

Age is not one scalar unless an identity class and clock are specified. Universopedia records

$$\tau = (\tau_{\text{signal}}, \tau_{\text{formation}}, \tau_{\text{material}}, \tau_{\text{processing}}). \tag{22.1}$$

Signal age
: propagation delay from emission or event to the record.

Formation age
: time since assembly of the selected effective object.

Material age
: distribution of ages of constituent matter or nuclei.

Processing age
: time since the last substantial local reconfiguration.

The four clocks can differ by orders of magnitude. A present-day planetary atmosphere contains nuclei older than the planet, has weather processed on short timescales, and is observed with a finite light-travel delay. Collapsing these into "the age" erases distinctions relevant to different questions.

22.2. When a collection becomes a field

For samples (x_i, z_i) and a scale-dependent kernel W_ℓ, define

$$F_\ell(x) = \frac{\sum_i W_\ell(x - x_i) f(z_i)}{\sum_i W_\ell(x - x_i)}, \qquad N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{22.2}$$

When $N_{\text{eff}} \approx 1$, the output remains effectively a local-object value. Field language becomes operationally justified when many contributions are involved, the result is stable under defensible changes of resolution, and the continuum approximation predicts the target observables within uncertainty.

22.3. A vector, not a single complexity essence

The corpus defines a scale-relative profile

$$\boldsymbol{\kappa} = (D, N, R, C, M, U), \tag{22.3}$$

for diversity, nesting, retention, channels, model depth, and uncertainty. The profile is descriptive. Averaging ordinal components can assist navigation but is not a law of nature, and uncertainty must not be hidden inside a single rank.

22.4. The periodic table as a complete language family

The periodic table is the corpus's principal example of a mature observational language: atomic identity is indexed by nuclear charge Z, refined by isotope A, ionization, electronic state, and environment. Spectral lines, isotope shifts, selection rules, and mass records allow the same atomic classes to be recognized across laboratories, stars, interstellar media, and engineered systems.

This does not make the periodic table a derivation from Unified Mechanics. It demonstrates what a successful familiarity system looks like:

1. stable class invariants;
2. multiple independent measurement channels;
3. environment-dependent subclasses;
4. explicit transmission limits such as opacity, blending, resolution, and selection rules;
5. conservative extension when a new isotope, state, or line class is learned.

The complete element-by-element ledger is preserved in Volume V's source and data crosswalk rather than repeated as foundational physics.

CHAPTER 23

Traversal tests and honest limits

23.1. A retained-fraction threshold

The corpus proposes r^3 as the loss assigned to a three-stage traversal. A possible optical test converts it into a retained fidelity threshold

$$\mathscr{F} \geq 1 - r^3 \approx 0.9704915028. \tag{23.1}$$

If expressed through a declared optical metric such as Strehl ratio, the metric and acceptance rule must be fixed before data are examined. The number is a forecast only after the physical mapping from traversal stages to the measured system is fixed.

FORECAST PROGRAMME 8.1 — PARAMETER-FREE TRAVERSAL RETENTION

Choose a physical channel with exactly three independently justified stages, specify a state-distance or wavefront metric, register Equation (23.1) and all analysis choices, and test on withheld data. A stage count selected after observing the loss would make the result a postdiction.

23.2. What the traversal programme establishes

TABLE 23.1.. Traversal and observation ledger.

Statement	Grade	Scientific boundary
Projected recurrence can hide state change	exact mathematics	relevance depends on future observables
Helical lift removes base recurrence	exact construction	physical depth is not yet derived
Causal order constrains conformal geometry	established	requires causal and regularity hypotheses
Gödel spacetime contains closed timelike curves	established	not experimentally realised by the corpus

Statement	Grade	Scientific boundary
Memory-erasing recurrence breaks record-defined identity	conditional interpretation	does not invalidate the spacetime solution
Holographic scale can be written as Hamilton–Jacobi flow	established in its domain	new history coordinate remains conjectural
Assembly requires relations as well as components	exact factorisation result	target-relative
Observation is mechanism/channel/record/inference chain	methodological standard	each application needs a concrete model
Transmission and familiarity kernels differ	exact conceptual distinction	familiarity is classifier-relative
r^3 traversal threshold	forecast proposal	needs preregistered stage and metric map

23.3. Forward link

Volume IV applies these rules to the corpus's physical claims. It reports established group-theoretic reproductions, finite sampling results, cosmological comparisons, simulation programmes, and null tests. Every empirical section begins with the measurement map and ends with the unresolved kernel.

PART IV

Phenomenology, Cosmology, and Physical Applications

CHAPTER 24

The contract for physical application

24.1. From formal structure to measured physics

The preceding volumes established a language for distinctions, a carrier geometry, and a disciplined account of traversal and readout. None of those achievements, by itself, identifies an electron, a galaxy, or a cosmological parameter. A physical application requires a realisation map

$$\mathscr{R} : \mathcal{S}_{\text{formal}} \longrightarrow \mathcal{S}_{\text{physical}} \tag{24.1}$$

and a measurement map

$$\mathscr{M} : \mathcal{S}_{\text{physical}} \longrightarrow \mathscr{D} \tag{24.2}$$

to calibrated data. An algebraic identity may be exact in $\mathcal{S}_{\text{formal}}$ while its proposed image under $\mathscr{R}$ is false. Numerical proximity cannot supply the missing map.

This volume therefore uses five evidential types.

TABLE 24.1.. Application-level evidential types.

Type	Meaning
Established	Published mathematics or experiment, with its actual hypotheses and uncertainty.
Internal result	Reproducible consequence of the corpus's declared definitions or data.
Conditional identification	Consequence obtained only after a stated formal object is mapped to a physical one.
Forecast	A predeclared numerical or qualitative output that later data can test.
Exposure	A missing derivation, non-identifiability, resolution boundary, or observation that can discriminate among or exclude a declared realization.

24.2. No scientific freezes

Earlier corpus documents sometimes used *freeze* to mean that a formula, convention, or scoring rule should not be altered while a particular audit was running. That is a valid preregistration device, but it cannot confer truth. In this edition every claim remains revisable. A later version may supersede an equation, mapping, or grade when a proof fails, a convention changes, or evidence improves. The earlier state remains in the provenance ledger so that the change is inspectable.

Words such as "exact", "forced", and "parameter-free" always have a domain. "Exact" means exact under the displayed definitions; "forced" names the assumptions that do the forcing; and "parameter-free" does not mean assumption-free, convention-free, or selected independently of the comparison.

24.3. The application sequence

The order is deliberately chronological in logical dependence. Matter representations come first because later atomic and cosmological identifications presuppose a particle language. Atomic structure follows as a finite, controlled spectral problem. Classical theories are then treated as limits and translations. Cosmology comes last because it combines geometry, matter, radiative transfer, inference, and survey selection in one measurement chain.

CHAPTER 25

Matter representations and the Standard Model contact

25.1. What is inherited from established representation theory

The carrier analysis of Volume II reaches the conditional reduction

$$\mathfrak{e}_8 \supset \mathfrak{so}(10) \oplus \mathfrak{su}(4), \tag{25.1}$$

equivalently the $D_5 \oplus A_3$ maximal-rank branching. At the level of the adjoint representation,

$$\mathbf{248} = (\mathbf{45}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{15}) \oplus (\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}}) \oplus (\mathbf{10}, \mathbf{6}). \tag{25.2}$$

This is standard group theory, not a new particle prediction. Grand-unified embeddings of one Standard Model generation in the spinor **16** of Spin(10), and its decomposition under $SU(5)$, have a long history (Georgi and Glashow, 1974; Fritzsch and Minkowski, 1975). Moreover, an E_8 representation by itself does not yield the observed chiral Standard Model; representation-theoretic obstructions must be addressed rather than hidden (Distler and Garibaldi, 2010).

The useful conventional identity is

$$\mathbf{16} \downarrow_{SU(5)} = \mathbf{10} \oplus \overline{\mathbf{5}} \oplus \mathbf{1}, \tag{25.3}$$

which is also realised by the even exterior algebra of five complex directions. Equation (25.3) packages one left-handed Weyl generation, including a sterile right-handed-neutrino conjugate.

25.2. Hypercharge and electric charge

Split the five complex directions into three colour directions and two weak directions. A generator constant on each block and traceless on the five-dimensional fundamental has, up to an overall normalization,

$$Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \tag{25.4}$$

Exterior products add the diagonal charges, and electric charge is

$$Q = T_3 + Y. \tag{25.5}$$

The resulting multiplets agree with the conventional one-generation charge table (Navas et al., 2024).

Table 25.1.. One generation written as left-handed Weyl fields.

Field	$SU(3)_c$	$SU(2)_L$	Y
Q_L	$\mathbf{3}$	$\mathbf{2}$	$1/6$
u_L^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	$-2/3$
d_L^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	$1/3$
L_L	$\mathbf{1}$	$\mathbf{2}$	$-1/2$
e_L^c	$\mathbf{1}$	$\mathbf{1}$	1
ν_L^c	$\mathbf{1}$	$\mathbf{1}$	0

Established result 2.1 — charge reconstruction

Given the conventional $SU(5)$ block split and the normalization in Equation (25.4), the Standard Model hypercharges and electric charges follow. The fractional quark charges reflect tracelessness across three colour and two weak directions. This reconstruction does not explain why nature selects that embedding, why there are three families, or how symmetry breaking occurs.

The same normalization gives the familiar unified boundary value

$$\sin^2 \theta_W = \frac{3}{8}. \tag{25.6}$$

Equation (25.6) is a high-scale group-normalization relation, not the measured low-energy weak mixing angle. Renormalization-group evolution, thresholds, and the symmetry-breaking spectrum are required before comparison with laboratory data.

25.3. Anomaly cancellation is a necessary arithmetic check

For the fields in the table, the mixed and Abelian gauge-anomaly sums vanish:

$$[SU(3)]^2U(1):\quad 2\left(\frac{1}{2}\right)\left(\frac{1}{6}\right)+\frac{1}{2}\left(-\frac{2}{3}\right)+\frac{1}{2}\left(\frac{1}{3}\right)=0, \tag{25.7}$$

$$[SU(2)]^2U(1):\quad 3\left(\frac{1}{2}\right)\left(\frac{1}{6}\right)+\frac{1}{2}\left(-\frac{1}{2}\right)=0, \tag{25.8}$$

$$[\mathrm{grav}]^2U(1):\quad 6\left(\frac{1}{6}\right)+3\left(-\frac{2}{3}\right)+3\left(\frac{1}{3}\right)+2\left(-\frac{1}{2}\right)+1=0, \tag{25.9}$$

$$[U(1)]^3:\quad 6\left(\frac{1}{6}\right)^3+3\left(-\frac{2}{3}\right)^3+3\left(\frac{1}{3}\right)^3+2\left(-\frac{1}{2}\right)^3+1=0. \tag{25.10}$$

The purely non-Abelian local cubic anomaly vanishes for $SU(2)$, while the even number of weak doublets avoids the global $SU(2)$ anomaly. These cancellations validate the reproduced charge assignment. They do not independently derive it.

25.4. The proposed matter-action square

The corpus extends the gravitational square of Volume II from the geometry block to the full carrier. Its three algebraic weights are

$$W_\mathrm{L}=u^2,\qquad W_\mathrm{B}=2ur,\qquad W_\mathrm{M}=r^2,\qquad W_\mathrm{L}+W_\mathrm{B}+W_\mathrm{M}=1, \tag{25.11}$$

where

$$r=\frac{1}{2\varphi}=0.309016994\ldots,\qquad u=1-r. \tag{25.12}$$

It associates the three terms of the square with gauge, matter-coupling, and mass sectors. Internally,

$$\frac{W_\mathrm{M}}{W_\mathrm{L}}=\frac{1}{5} \tag{25.13}$$

is exact. Physically, however, an action is more than three coefficients: it requires fields, representations, covariant derivatives, kinetic terms, symmetry breaking, reality conditions, and equations of motion.

Conditional identification 2.1 — the matter square

If the carrier projection selects the conventional gauge and spinor modules and if the three terms of Equation (25.11) multiply correctly normalized gauge, fermion, and mass operators, then their relative coefficients are fixed by r. Those operators and their normalization have not yet been derived, so no Standard Model coupling or particle mass follows at present.

25.5. Chirality, mass, and the strong-CP boundary

The corpus proposes that reversible carrier directions support visible gauge transport, while an irreversible record condition protects chirality by excluding a traversal immediately composed with its reverse. This is a structural picture, not yet a quantum field theory. A finite decay group alone does not evade the representation-theoretic obstruction identified for four-dimensional E_8 unification (Distler and Garibaldi, 2010); a concrete chiral spectrum and anomaly-free action are required.

Likewise, identifying colour-block reversibility with a vanishing QCD vacuum angle does not calculate the strong-CP parameter. The standard strong-CP problem and its Peccei–Quinn/axion solution have well-defined field-theoretic content (Peccei and Quinn, 1977; Weinberg, 1978; Wilczek, 1978), and neutron electric-dipole searches constrain CP violation directly (Abel et al., 2020).

Exposure 2.1 — the decisive matter calculation

Construct a local, unitary, anomaly-free action; derive its chiral fermion content, gauge couplings, Yukawa matrices, symmetry-breaking pattern, and strong-CP sector; and run its parameters to measured scales. Charge-table reproduction and the coefficient ratio $1/5$ do not substitute for this calculation.

CHAPTER 26

Hydrogen, finite designs, and atomic structure

26.1. The established hydrogen symmetry

The nonrelativistic Coulomb Hamiltonian

$$H = \frac{\mathbf{p}^2}{2\mu} - \frac{Ze^2}{4\pi\varepsilon_0 r} \tag{26.1}$$

has, for bound states, the familiar energy spectrum

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}, \qquad n = 1, 2, \ldots . \tag{26.2}$$

The angular momentum and Runge–Lenz vector generate an $SO(4)$ dynamical symmetry on a fixed negative-energy shell. Fock's momentum-space construction maps the bound-state integral equation to a problem on S^3 (Fock, 1935). The momentum sphere and Equation (26.2) are established quantum mechanics; they cannot be counted as predictions of the carrier framework.

26.2. What the discrete polytopes actually establish

A finite set $X \subset S^3$ is a spherical t-design when averaging every polynomial of total degree at most t over X reproduces the spherical average (Delsarte et al., 1977). The 24-cell vertices form a 5-design, and the 120 vertices of the 600-cell form an 11-design (Coxeter, 1973; Conway and Sloane, 1999). Because a product of harmonics of degree at most ℓ has degree at most 2ℓ, a t-design exactly reproduces the corresponding inner products when

$$2\ell \le t. \tag{26.3}$$

Hydrogenic states with principal quantum number n contain $\ell \le n - 1$. Therefore

design	t	exact harmonic sector
24-cell	5	$\ell \le 2 \quad (n \le 3)$,
600-cell	11	$\ell \le 5 \quad (n \le 6)$.

(26.4)

Internal result 3.1 — finite quadrature cutoffs

Equation (26.4) follows from the published design strengths and the degree of harmonic products. It validates exact finite quadrature for the stated sectors. It does not derive the Coulomb Hamiltonian, the Rydberg scale, relativistic fine structure, Lamb shifts, or transition rates.

The earlier corpus route that inferred the hydrogen spectrum from a carrier S^3 was circular: it used the same momentum-space symmetry structure that is obtained only after the Coulomb problem and its energy shell have been supplied. Self-duality of a lattice does not identify position and momentum spaces as one physical manifold without a symplectic/Fourier realisation. The scientifically retained result is finite sampling, not an independent derivation of hydrogen.

26.3. Atomic and chemical data as an observational atlas

The Atomic Web material records all 118 elements with identifiers, display data, and audit files. It is valuable as a provenance-controlled atlas and as an example of the Universopedia observational standard. The element classes themselves come from established nuclear charge, isotope, electronic-structure, and spectroscopy data; the colour web is a visualization, not a new periodic law.

Exposure 3.1 — from quadrature to atomic physics

A genuine carrier prediction must begin with a derived interaction and Hilbert space, then return at least one previously uninserted energy, degeneracy breaking, transition amplitude, or selection rule. Matching a known S^3 representation or sampling its harmonics is insufficient.

CHAPTER 27

Classical theories as limits and translations

27.1. Electromagnetism

The established field theory is generated by

$$S_{\mathrm{EM}} = -\frac{1}{4}\int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x + \int A_\mu J^\mu \sqrt{-g}\,\mathrm{d}^4x, \tag{27.1}$$

with $F = \mathrm{d}A$, gauge redundancy $A \mapsto A+\mathrm{d}\chi$, and charge conservation following from the source coupling and gauge symmetry. The corpus's Light/Boundary/-Matter vocabulary can label propagation, connection, and retained source structure, but relabelling Maxwell theory is not a derivation. A new electromagnetic sector would need a distinct constitutive relation, coupling, dispersion law, or boundary prediction.

27.2. Thermodynamics and statistical mechanics

The corpus interprets entropy increase as growth of distinctions that cannot be harmlessly overwritten in a chosen record description. This is compatible with coarse-graining, but it must not be confused with the microscopic origin of thermodynamic irreversibility. For a distribution p_i,

$$S = -k_{\mathrm{B}}\sum_i p_i \ln p_i \tag{27.2}$$

is a state functional once the ensemble and coarse-graining are specified. Reversible microscopic equations, typicality, environmental interaction, initial conditions, and the selected macrovariables determine when a macroscopic entropy law follows.

Shannon entropy quantifies uncertainty in a coding distribution (Shannon, 1948); Landauer's bound concerns logically irreversible erasure implemented physically (Landauer, 1961; Bennett, 1982). Neither result proves a universal metaphysical prohibition on erasing records. The strongest corpus statement is target-relative: erasing a distinction that later changes an admissible readout makes the compressed state dynamically insufficient.

27.3. The status of the classical manuscripts

The legacy electromagnetism, thermodynamics, and statistical-mechanics texts are therefore retained as interpretive translations and consistency checks. Where they reproduce standard equations, the equations keep their standard evidential status and literature. Where they add only new names, no independent confirmation is claimed. Where they propose a new record-dependent constitutive law, that law enters the exposure ledger until an experiment is specified.

CHAPTER 28

Cosmological baseline and algebraic dictionary

28.1. The standard background

For a spatially homogeneous and isotropic universe, the Friedmann equations are

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{28.1}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \tag{28.2}$$

with component conservation

$$\dot{\rho}_i + 3H\left(\rho_i + \frac{p_i}{c^2}\right) = 0. \tag{28.3}$$

These equations follow from general relativity under the symmetry assumptions (Einstein, 1916; Friedmann, 1922). They define background expansion; they do not by themselves calculate recombination, the primordial spectrum, perturbation transfer, galaxy bias, or a dark-matter particle.

Planck's six-parameter flat-ΛCDM analysis gives, among other values, $n_s = 0.9649 \pm 0.0042$, $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and matter density $\Omega_m \simeq 0.315$ for the stated data combination and model (Planck Collaboration, 2020b). The SHoES baseline distance ladder reported $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Riess et al., 2022). These are model- and pipeline-conditioned inferences, not direct readings of a single instrument dial.

28.2. The fixed partition

From the formal update in Volume I,

$$r = \frac{1}{2\varphi}, \qquad u = 1 - r, \tag{28.4}$$

and the square of the two channels gives

$$W_{\mathrm{L}} = u^2 = 0.477457514063\dots, \tag{28.5}$$

$$W_{\mathrm{B}} = 2ur = 0.427050983125\dots, \tag{28.6}$$

$$W_{\mathrm{M}} = r^2 = 0.095491502813\dots. \tag{28.7}$$

The partition and identities such as $W_{\mathrm{L}}/W_{\mathrm{M}} = 5$ are exact. The cosmological dictionary is a later hypothesis:

$$\Omega_b^{\mathrm{UM}} = \frac{W_{\mathrm{M}}}{2} = 0.0477457514\dots, \tag{28.8}$$

$$\Omega_c^{\mathrm{UM}} = \frac{W_{\mathrm{B}}}{\varphi} = 4r^2u = 0.2639320225\dots, \tag{28.9}$$

$$\Omega_{\mathrm{DE}}^{\mathrm{UM}} = 1 - \Omega_b^{\mathrm{UM}} - \Omega_c^{\mathrm{UM}} = 0.6883222261\dots, \tag{28.10}$$

$$Y_{\mathrm{He}}^{\mathrm{UM}} = \frac{W_{\mathrm{L}}}{2} = 0.2387287570\dots, \tag{28.11}$$

$$N_{\mathrm{eff}}^{\mathrm{UM}} = 3 + \frac{W_{\mathrm{M}}}{2} = 3.0477457514\dots, \tag{28.12}$$

$$n_s^{\mathrm{dict}} = 1 - W_{\mathrm{M}}(1-2r) = 0.9635254916\dots, \tag{28.13}$$

$$\tau^{\mathrm{dict}} = 2r^3 = 0.0590169944\dots. \tag{28.14}$$

CONDITIONAL IDENTIFICATION 5.1 — COSMOLOGICAL DICTIONARY

Every number in Equation (28.14) is an exact algebraic output of the declared map. The map is not derived by the partition theorem, and several entries are standard-fit parameters with correlated likelihoods rather than independent observables. Agreement after the dictionary was proposed is retrospective and cannot be graded as a prediction.

28.3. The evolved readout and its selection cost

The Field Ledger later evolves the fixed pole, assigns a phase, uses helical turn count as position along the path, and weights readings by a resolution kernel. In compact form,

$$\mathscr{R}[O] = \frac{\sum_t H_5(N_t)O(C_t)}{\sum_t H_5(N_t)}, \qquad N_t = \frac{360T_O}{\Delta\theta_t}, \qquad H_5(N) = \frac{2s^{10}}{1+s^{10}}, \quad s = \frac{N}{N+1}. \tag{28.15}$$

The implementation reproduces the reported table and moves all six selected residuals toward the chosen Planck comparison. That is a reproducible property of the algorithm. It is not an out-of-sample likelihood result: the observable set, channel typing, optical-depth path factor, and comparison target were already in contact with the data.

INTERNAL RESULT 5.1 — DETERMINISTIC RECONSTRUCTION

Given the declared evolution, phase assignment, turn count, and kernel in Equation (28.15), the corpus's six-entry table regenerates deterministically. Its mean absolute standardized residual decreases from 0.9979 to 0.6713. This is an in-sample reconstruction, not evidence with a calculable sampling significance.

The former "traversal payment"

$$P_T = 1 - \frac{E_1}{E_0} = 0.327356 \tag{28.16}$$

is consequently a score-improvement statistic for that typed audit. Its near-one-third value is not a universal constant, and the later typing ladder correctly reinterprets it as dependent on which records have been classified. Nothing about this score is frozen in the living edition.

CHAPTER 29

The rate across cosmological scale

29.1. An exact mixture identity

Let $N = \ln a$ and define the total acceleration parameter

$$\epsilon = -\frac{\mathrm{d}\ln H}{\mathrm{d}N}. \tag{29.1}$$

For noninteracting components with constant equations of state w_i, write

$$\epsilon_i = \frac{3}{2}(1 + w_i), \qquad \epsilon = \sum_i f_i \epsilon_i, \tag{29.2}$$

where $f_i = \rho_i/\rho$. Since

$$\frac{\mathrm{d}f_i}{\mathrm{d}N} = 2 f_i(\epsilon - \epsilon_i), \tag{29.3}$$

one obtains

$$\frac{\mathrm{d}\epsilon}{\mathrm{d}N} = 2\left(\epsilon^2 - \langle\epsilon_i^2\rangle\right) = -2\,\mathrm{Var}_f(\epsilon_i) \leq 0. \tag{29.4}$$

If the Hamilton–Jacobi curvature parameter is defined by

$$\frac{\mathrm{d}\epsilon}{\mathrm{d}N} = 2\epsilon(\epsilon - \eta_H), \tag{29.5}$$

then, whenever $\epsilon \neq 0$,

$$\eta_H = \frac{\langle\epsilon_i^2\rangle}{\langle\epsilon_i\rangle}. \tag{29.6}$$

Equation (29.5) is standard Hamilton–Jacobi flow (Salopek and Bond, 1990; Skenderis and Townsend, 2006); Equation (29.6) is its fluid-mixture rewriting under the assumptions just stated.

A cosmological constant has $w = -1$ and $\epsilon_\Lambda = 0$, so its direct contribution cancels from numerator and denominator of Equation (29.6). For pressureless matter plus Λ,

$$\eta_H = \frac{3}{2}, \qquad \epsilon(z) = \frac{3}{2}\frac{\Omega_m(1+z)^3}{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{29.7}$$

ESTABLISHED RESULT 6.1 — WHAT CONSTANT η_H MEANS LATE

Equation (29.7) exactly rewrites the flat matter-plus-Λ background as the constant-$\eta_H = 3/2$ trajectory. This is a useful invariant representation of ΛCDM, not an independent derivation or confirmation of it. Radiation, curvature, interactions, and evolving equations of state modify the mixture.

29.2. Inflationary tilt and running

For single-field inflation in the Hubble slow-roll expansion, to first order,

$$n_s - 1 \simeq 2\eta_H - 4\epsilon, \qquad r_{\rm tens} \simeq 16\epsilon. \tag{29.8}$$

Here $r_{\rm tens}$ denotes the tensor-to-scalar ratio and must not be confused with the corpus retention fraction r. Planck found $n_s = 0.9649 \pm 0.0042$ and no evidence for running at its sensitivity (Planck Collaboration, 2020a); BICEP/Keck obtained $r_{0.05} < 0.036$ at 95% confidence from the 2018 observing-season analysis (BICEP/Keck Collaboration, 2021).

If η_H is treated as constant, differentiating the first-order relation gives, at the corresponding order,

$$\alpha_s \equiv \frac{\mathrm{d}n_s}{\mathrm{d}\ln k} \simeq -8\epsilon(\epsilon - \eta_H) \simeq -\frac{r_{\rm tens}(1-n_s)}{4}, \tag{29.9}$$

where $\mathrm{d}\ln k = (1-\epsilon)\mathrm{d}N$ and higher-order slow-roll terms have been neglected. The inherited manuscript described Equation (29.9) too strongly. It is not exact: both the spectral-index relation and the conversion between N and $\ln k$ are approximations at this order.

FORECAST 6.1 — CONSTANT-CURVATURE INFLATION

Under single-field Hubble slow roll with approximately constant η_H, Equation (29.9) predicts negative running whose magnitude is below a few 10^{-4} when the quoted n_s and tensor bound are used. A robust positive running, or a magnitude incompatible with the complete higher-order constant-η_H calculation, would reject this condition.

29.3. The proposed retention map

The framework separately proposes that one carrier closure corresponds to one unit of logarithmic cosmological scale. Fixed retention would then give a constant exponent,

$$n_s^{\rm flow} = 1 + \ln\big(1 - r^3\big) = 0.9700474\ldots. \tag{29.10}$$

The arithmetic is exact, but the identification of a carrier closure with an e-fold or a mode-label interval is not derived. Equation (29.10) is a postdiction because n_s was known when the map was proposed.

Exposure 6.1 — closure-to-scale bridge

Derive the map from carrier closure count to comoving wavenumber, including normalization and perturbation dynamics, without using the observed tilt. Until then, the closeness of Equation (29.10) to Planck is numerology with a clearly stated research target, not a primordial-spectrum calculation.

CHAPTER 30

Dark matter, galaxies, and structure formation

30.1. Density identity versus physical substance

The exact internal identity

$$\Omega_c^{\mathrm{UM}} = \frac{W_{\mathrm{B}}}{\varphi} = 4r^2(1-r) = 0.2639320225\ldots \tag{30.1}$$

is numerically close to the cold-dark-matter density inferred in flat ΛCDM fits. It does not specify what clusters. A viable dark sector must supply a stress–energy tensor and, at minimum, the background and linear behavior

$$\dot{\rho}_c + 3H\rho_c = 0, \tag{30.2}$$

$$\ddot{\delta}_c + 2H\dot{\delta}_c - 4\pi G\rho_m\delta_m \simeq 0, \tag{30.3}$$

$$w_c \simeq 0, \qquad c_s^2 \simeq 0, \qquad Q_{\mathrm{visible}} = 0. \tag{30.4}$$

The corpus proposes self-holding carrier modes that are neutral under the visible gauge projection. No mode operator, abundance calculation, perturbation closure, or nonlinear solution is yet available.

RESEARCH TARGET 7.1 — NEUTRAL SELF-HOLDING MODES

Derive a stable neutral spectrum whose stress–energy satisfies Equation (30.4); then predict linear transfer functions, halo profiles, substructure, lensing, and interactions from the same parameters. The numerical density identity is not admissible as a substitute for these outputs.

The earlier "vacuum crowding" mechanism is withdrawn. In the corpus's own galactic estimate it is about eleven orders of magnitude too small and scales like vacuum density times a weak gravitational potential. The calculation remains in the provenance ledger precisely to prevent its accidental revival under a new label.

30.2. Simulation and galaxy ledgers

IllustrisTNG provides a public suite of magnetohydrodynamic cosmological simulations with subgrid galaxy-formation prescriptions (Springel et al., 2018; Nelson et al., 2019). SPARC provides 3.6-μm photometry and high-quality rotation curves for 175 disk galaxies (Lelli et al., 2016); these data exhibit a tight radial-acceleration relation between observed and baryonic accelerations (McGaugh et al., 2016). Both are valuable test surfaces, but they answer different questions.

The corpus's TNG, density-gradient-flow, catalogue, planet-hunt, and age-matching directories are exploratory programmes and data-engineering records. They demonstrate pipelines, candidate filters, or possible signatures; they do not document a detection of a new planet, field, or dark-matter law. A post hoc fit to a galaxy catalogue is especially vulnerable to degeneracy among stellar mass-to-light ratio, gas distribution, distance, inclination, feedback, halo response, and selection.

30.3. A fair discriminating programme

A useful confrontation should proceed in this order:

1. derive the field or particle equations before selecting target galaxies;
2. calibrate nuisance parameters on a declared training set;
3. publish withheld-galaxy predictions for rotation, lensing, and where possible vertical dynamics;
4. compare against baryon-plus-halo and modified-dynamics baselines with the same nuisance freedom;
5. report failures, covariance, and selection functions as prominently as favorable cases.

Exposure 7.1 — dark-sector defeat condition

If the derived neutral modes cannot reproduce the CMB transfer function, large-scale growth, galaxy and cluster lensing, and nonlinear structure with one consistent stress–energy model, that declared dark-matter realization is empirically excluded even though Equation (30.1) remains an algebraic identity.

CHAPTER 31

The microwave background and expansion history

31.1. What the existing forward solver computes

The corpus solver uses Saha ionization equilibrium to estimate recombination,

$$\frac{x_e^2}{1-x_e} = \frac{1}{n_H}\left(\frac{m_e k_\mathrm{B} T}{2\pi\hbar^2}\right)^{3/2} \exp\!\left(-\frac{\chi_H}{k_\mathrm{B}T}\right), \tag{31.1}$$

then integrates the sound horizon and comoving angular-diameter distance,

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}, \qquad c_s = \frac{c}{\sqrt{3(1+R)}}, quad R = \frac{3\rho_b}{4\rho_\gamma}. \tag{31.2}$$

With seven invariant readouts and the observed background temperature, it returns $H_0 = 66.965\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ after imposing its acoustic-angle condition. It also returns $z_* \simeq 1369$, far earlier than the realistic last-scattering value near 1090. The resulting acoustic peak locations are correspondingly 13–19% high.

Internal result 8.1 — limited background solver

The reported numbers reproduce within the Saha-based code. The program calculates background distances and approximate scales; it is not a CMB likelihood or Boltzmann solver. The common peak displacement is a useful diagnostic of inadequate recombination, not a successful spectrum prediction.

Superseding computation 8.2 — full transfer evolution

The Nine-Step Closure performs the required replacement using CAMB 2.0.1: non-equilibrium recombination and the linear Einstein–Boltzmann hierarchy are evolved with the declared Unified Mechanics boundary realization. The run gives $z_* = 1089.746$, $r_* = 143.971\,\mathrm{Mpc}$, $H_0 = 67.753\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from the frozen acoustic anchor, and temperature peaks at $\ell = 220, 536, 813, 1126$ (Shields, 2026). The remaining cosmological work is a precision public-likelihood evaluation, an independent HyRec/CosmoRec cross-run, and completion of the discrete microscopic matter identity. The earlier Saha calculation remains provenance for the route by which the complete transfer calculation was identified.

31.2. The two-basin interpretation of H_0

Starting from the solver value $H_0^{\mathrm{low}} = 66.965$, the corpus applies three closure conventions, including

$$H_0^{\mathrm{high}} = H_0^{\mathrm{low}}(1 - r^3)^{-3} = 73.261\ldots. \tag{31.3}$$

Nearby variants give 72.893 and 73.168. The arithmetic is exact, and the transformed values are close to the SH0ES baseline. But there is no derived observable map determining which experimental pipeline receives which closure count.

Conditional interpretation 8.1 — expansion basins

If early- and late-universe distance indicators couple to distinct states of one scale-dependent field, then different inferred H_0 values need not estimate the same effective quantity. Equation (31.3) does not establish that premise. The basin-selection functional must be derived and applied prospectively to strong lenses, standard sirens, megamasers, red-giant distances, and inverse ladders.

31.3. Dark energy after DESI DR2

DESI DR2 baryon-acoustic-oscillation measurements are consistent with a flat ΛCDM description of the BAO data. When combined with CMB and supernova samples, the inferred preference for evolving dark energy depends on the supernova compilation and model combination (DESI Collaboration, 2025). It is therefore scientifically incorrect to describe evolution as a settled single-survey detection.

The corpus proposes that a fixed geometric vacuum record has exactly $w = -1$, while apparent evolution arises when readouts with different traversal counts are combined. This is a testable interpretation only after the traversal-to-likelihood map is specified. A physical equation of state cannot be dismissed merely because different supernova pipelines return different posterior contours; calibration and population systematics must be modelled quantitatively.

Forecast 8.1 — static-vacuum pipeline test

Derive, before a later survey combination is examined, the predicted $H(z)$, $D_M(z)$, $f\sigma_8(z)$, lensing response, and pipeline-dependent bias for the proposed fixed-vacuum model. A statistically compelling evolving equation of state in a single appropriately modelled BAO/growth pipeline, incompatible with the derived bias, would reject the static-vacuum interpretation.

CHAPTER 32

Vacuum scale, gravity tests, and the experimental ledger

32.1. The dimensionless vacuum readout

The corpus defines

$$\lambda_{\mathrm{UM}} = \frac{4\pi}{\sqrt{3}} r^{240} = 2.860333 \times 10^{-122}, \tag{32.1}$$

using the 240 roots of E_8, a triangular cell factor, and a genus-zero boundary factor. Equation (32.1) is an exact number after those choices. Identifying it with the observed dimensionless cosmological constant is highly convention-sensitive: the manuscript selects the non-reduced Planck length, and the reduced convention changes the dimensionless normalization by powers of 8π. The root-count exponent and geometric prefactor were also proposed after the vacuum hierarchy was known.

Conditional identification 9.1 — vacuum number

If each of 240 carrier crossings contributes one factor of r, if the observer boundary is genus zero, if the triangular cell and Gauss–Bonnet normalizations are physical, and if the non-reduced Planck convention is the correct observational dictionary, then Equation (32.1) maps to the vacuum term. The numerical match is a postdiction with several load-bearing conditions.

The associated dimensional construction

$$a_0 = \left(\frac{\hbar c}{\rho_\Lambda}\right)^{1/4} \simeq 88.1\,\mu\mathrm{m} \tag{32.2}$$

is the fourth-root length of a vacuum energy density. The relation is dimensional; interpreting a_0 or a weighted branch of it as a carrier-cell size is the new claim. A viable short-range-gravity forecast must specify the modified potential, coupling strength, composition dependence, and branch before comparison.

32.2. Equivalence and post-Newtonian gravity

The helical metric compression of Volume II uses

$$A = -\alpha_M, \qquad B = \alpha_L + \frac{\nu}{3}. \tag{32.3}$$

Imposing $\nu = 3(\alpha_M - \alpha_L)$ makes the post-Newtonian slip

$$\gamma = \frac{\alpha_L + \nu/3}{\alpha_M} = 1 \tag{32.4}$$

identically. This proves that the chosen map cannot produce slip; it does not derive the map or its sources. Cassini constrained $\gamma - 1$ through radio tracking near solar conjunction (Bertotti et al., 2003), and MICROSCOPE found no weak-equivalence-principle violation at its final sensitivity (Touboul et al., 2022). Any helical field must pass these results together with laboratory, binary-pulsar, lensing, and gravitational-wave constraints.

The internal identity

$$(r\varphi)^2 = \frac{1}{4} \tag{32.5}$$

has the same coefficient as the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A/(4\ell_P^2)$ in natural units (Bekenstein, 1973; Hawking, 1975). Equality of coefficients is a contact, not a microstate count or a derivation of horizon thermodynamics.

32.3. A chronological discrimination ledger

TABLE 32.1.. Physical sector: current result and next discriminating calculation or test.

Sector	Present status	Next discriminating calculation or test
Standard Model charges	chiral 16, charges, anomaly sums, and one-loop coefficients fixed	stationary mass selector, threshold-matched running, and mixing matrices from one pre-data rule
Matter square	registered gravitational, gauge, and chiral variations; boundary mass-squared block located	discrete matter-to-cell constitutive identity and twelve-component stationary spectrum

Sector	Present status	Next discriminating calculation or test
Strong CP	reversibility interpretation only	calculate $\bar{\theta}$ and neutron EDM without inserting the observed bound
Hydrogen	exact finite-design quadrature through stated harmonic sectors	predict an uninserted atomic energy or transition observable
Cosmological dictionary	covariant vacuum and retained-matter tensors with linear perturbation equations	public multi-observable likelihood and microscopic constitutive closure
Constant η_H	exact fluid identity late; leading-order inflationary forecast	higher-order calculation conflicts with measured running/tensors
Dark matter	conserved dust-level effective tensor and perturbation system	one microscopic identity must reproduce CMB, growth, lensing, and nonlinear structure consistently
CMB solver	full CAMB recombination and Einstein–Boltzmann evolution completed	precision likelihood, polarization/lensing assessment, and independent recombination cross-run
Hubble basins	numerical transforms, selection rule absent	prospectively assigned intermediate probes do not occupy predicted basins
Static vacuum	conditional $w = -1$ interpretation	derived single-pipeline prediction is excluded by expansion and growth data
Vacuum number	convention-sensitive postdiction	physical normalization or preregistered short-range signal fails
Helical gravity	algebraic $\gamma = 1$ under imposed source relation	derived source action violates equivalence, post-Newtonian, lensing, pulsar, or wave tests

32.4. What this volume establishes

The physical chain is now continuous at the level carried by the corpus: the one-generation charge pattern is reproduced from the five-direction block split; the chiral matter and anomaly conditions are explicit; the local action yields its registered gravitational, gauge, and matter equations; finite polytopes supply exact quadrature in the stated hydrogenic sectors; conserved dark-sector tensors enter the linear perturbation hierarchy; and the cosmological boundary realization has been propagated through non-equilibrium recombination and full linear Einstein–Boltzmann evolution.

The remaining work is concentrated rather than diffuse. It consists of the twelve-component stationary mass solution, the discrete microscopic matter identity, the carrier threshold and normalization calculation, the short-range response kernel, precision likelihood work, and genuinely prospective measurement. Volume V and the Nine-Step Closure record the exact dependencies and certificates so those tasks extend the theory without confusing an active interface with an absent framework.

PART V

Formal Register, Methods, and Reproducibility

CHAPTER 33

The living scientific record

33.1. Purpose of the register

This volume is the audit surface for the series. It does not add another layer of physical ontology. It records the commitments on which the argument depends, the meanings of its terms, the scientific grade of each inherited theorem, the relation between source documents and rewritten chapters, and the conditions under which a result must be revised or abandoned.

A register is necessary because the corpus developed through books, technical papers, addenda, computational notebooks, generated tables, data files, manifests, and several editorial states. Later documents often corrected earlier ones, while exact duplicates appeared in different formats or directories. Preserving everything therefore cannot mean repeating every sentence. It means preserving each distinct claim, equation, dataset, counterexample, certificate, failure, and historical transition in a structure where its current disposition is visible.

33.2. No-freeze versioning

Scientific content is never frozen. At most, an analysis protocol may be preregistered for the duration of a test. The living edition separates four operations:

Version
: a citable state of the manuscript, code, data, and register;

Correction
: a change that repairs an error without changing the intended claim;

Supersession
: a new claim or derivation that replaces an earlier one while retaining its provenance;

Withdrawal
: a route that no longer supports the stated conclusion, retained so that it cannot silently return.

No manifest may declare a proposition scientifically immutable. Reproduction means that a result can be regenerated from a declared historical state; it does not require future editions to continue believing the result.

Editorial rule 1.1 — provenance without authority

Every superseded or withdrawn item remains addressable by source path and digest. Its survival in the record establishes what was written and tested, not that the claim is true. The current scientific grade is governed by the latest explicit regrading, which may itself later be superseded.

33.3. Corpus census

The combined audit covered the repository and the larger SCI archive. It indexed 1,319 files, representing 791 distinct SHA-256 hashes. Of these, 491 unique text-bearing sources were extracted into the claim-and-coverage ledger; 155 were classified as substantive scientific sources and 132 as manuscript files. Exact duplicate groups and cross-format renderings were consolidated, while their paths were retained. No text extraction failed in the final index.

The 491-source crosswalk contains 148 implementation files, 126 manuscripts, 105 data or certificate files, 64 supporting texts, 23 orientation or ledger files, 19 manifests, and 6 historical manuscripts. The machine-readable ledger records every source's full digest, format, role, approximate word count, leading topics, claim labels, assigned volume and section, and review status.

CHAPTER 34

Scientific grades and inference rules

34.1. A typed claim system

The inherited corpus frequently used PROVED for several logically different achievements. This edition replaces that single word with a typed system.

TABLE 34.1.. Scientific grades used throughout the living edition.

Grade	Criterion
Definition	Fixes vocabulary or a mathematical object. It is neither true nor false until judged for consistency and usefulness.
Framework postulate	A declared starting commitment. Consequences remain conditional on it.
Exact identity	Follows by algebra from displayed definitions, with no physical identification implied.
Internal lemma/theorem	Follows from stated formal hypotheses by a checkable proof.
Computational result	Regenerated by a declared algorithm and certificate; scope is limited by finite precision, enumeration completeness, and implementation assumptions.
Established external result	A published theorem or observation imported with its actual hypotheses, model, units, and uncertainty.
Conditional identification	Connects an internal object with a physical one only after a named realisation map is granted.
Interpretation	A coherent reading that does not itself add a testable dynamical consequence.
Conjecture	A mathematical or physical statement for which the required proof or mechanism is absent.
Forecast	A predeclared output exposed to future or withheld data.
Open obligation	A missing bridge whose completion is required by a stronger claim.

Grade	Criterion
Withdrawn	A retained historical route that no longer supports its conclusion.

34.2. The claim tuple

Each serious claim should be representable as

$$\mathfrak{C} = (S, H, D, M, R, G, F), \tag{34.1}$$

where S is the statement, H its hypotheses, D the derivation or data, M the measurement map, R the reproducible artefact, G the grade, and F a falsifier or supersession condition. A purely mathematical identity may leave M empty. A physical claim may not.

The following invalid moves recur in speculative theory and are explicitly prohibited:

1. promoting an exact identity to a physical law without a realisation map;
2. treating numerical agreement after model selection as a forecast;
3. using a standard result as evidence for the framework merely because the framework can restate it;
4. counting correlated parameters as independent confirmations;
5. declaring a missing mechanism "forced" by a verbal definition;
6. changing conventions, datasets, stage counts, or branches after viewing the residual without recording the selection;
7. interpreting failure to falsify as confirmation.

34.3. Dependency and proof discipline

For each theorem, the proof graph must distinguish four edge types:

$$\begin{gathered}\text{definition} \longrightarrow \text{formal consequence} \\ \longrightarrow \text{realisation assumption} \longrightarrow \text{empirical readout.}\end{gathered} \tag{34.2}$$

An external theorem enters only at the edge licensed by its hypotheses. For example, lattice classification can identify E_8 after rank, integrality, evenness, unimodularity, and positive definiteness have been supplied; it cannot supply those premises. Gleason's theorem constrains probability measures under non-contextual additivity hypotheses (Gleason, 1957; Busch, 2003; Caves et al., 2004); it does not derive those hypotheses from fairness language. Wigner and Stone constrain transformations only after transition-probability preservation, bijectivity, and continuity have been stated (Wigner, 1959; Stone, 1932).

CHAPTER 35

Commitments and definitions

35.1. Commitments A1–A6

The six inherited axioms are preserved below with their living-edition status.

Table 35.1.. Foundational commitments.

ID	Name and grade	Scientific statement
A1	Systemhood; framework postulate	A model may treat a system as one entity while permitting internally distinguishable states or parts. This is a modelling convention, not an empirical law.
A2	Ground; ontological postulate	The first distinction is said to constitute its own "where". No spacetime consequence follows until a mathematical state space and geometry are constructed.
A3	Two-sided reading; framework postulate	A mixed relation is counted in both orientations. This supplies the factor of two in the retained-fraction normalization; all downstream numerical claims remain conditional on it. It does not by itself prove evenness or lattice self-duality.
A4	Saturation; open physical postulate	A single accumulating channel is assumed to reach finite capacity. Capacity, threshold, state space, and dynamics remain unspecified.
A5	Conservation of readout; conservation postulate	Continuing state, outgoing record, and their order are assumed to conserve a declared total. Physical use requires an explicit conserved measure or current.

ID	Name and grade	Scientific statement
A6	Admissible reduction; definition	A reduction is declared unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. Operator-algebra consequences require a specified algebra, action, norm, and fixed space.

35.2. Definitions D1–D18

Table 35.2.. Framework vocabulary, preserved and delimited.

ID	Term	Living-edition definition
D1	Distinction	A relation or property that prevents complete equivalence under the comparison being made.
D2	Identity	Recoverability of declared distinctions through an admissible family of changes.
D3	Relation	A rule by which distinct contents compare, constrain, or transform one another without automatic identification.
D4	Seven doings	Distinguish, hold, close, answer, change, order, and record: the corpus's constitutive grammar.
D5	Traversal stages	Persist, distinguish, relate, propagate, resolve, return, and close: a process-oriented rendering of D4.
D6	Object	A stable equivalence class of closed traversals relative to a declared observation and timescale.
D7	Update	The two-channel map $(a, b) \mapsto (a + b, a)$.
D8	Channels	The outward share u and retained share r, normalized by $u + r = 1$.
D9	Three weights	$W_L = u^2$, $W_B = 2ur$, and $W_M = r^2$. The physical names are interpretations.
D10	Crossing	One enacted step of self-description across a declared channel. Its assignment of cost r is a separate conjectural rule.

ID	Term	Living-edition definition
D11	Record	A distinction that remains discriminable and recoverable under the future dynamics relevant to the task. The inherited binary criterion is an idealization; physical records can be noisy or graded.
D12	Observer	A system in which another system's distinction becomes coupled and persistently recorded; a mechanical observer includes preparation, coupling, propagation, resolution, amplification, and record retention.
D13	Carrier	The structure on which the framework writes self-describing records. Self-duality is a modelling requirement, not a derivation of an even unimodular lattice.
D14	Primitive crossing	A smallest nonzero displacement in a chosen carrier model; in the E_8 realisation these are roots of squared norm two.
D15	Cluster projection	A map C from carrier observables to a readable subalgebra satisfying A6.
D16	Decay generator	An operator K whose kernel is the readable algebra; the minimal choice is $I - C$, while a general positive generator may have a richer spectrum.
D17	Readable algebra	The subalgebra fixed by the declared reduction and therefore retained by the effective description. Its identification with physical observables is model-dependent.
D18	Geometry projectors	Orthogonal projectors P_X, P_T, P_I assigned to external distinction, causal order, and closure, with $P_X + P_T + P_I$ the geometry block. Their ranks are assumptions until represented independently.

CHAPTER 36

Regraded formal register T1–T39

The following table is the controlling scientific register. It preserves the inherited numbering so every earlier proof, certificate, and cross-reference remains traceable. The short ruling is not a replacement for the fuller derivations in Volumes I–IV; it states the maximum defensible grade.

TABLE 36.1.. Living-edition regrade of the inherited theorem register.

ID	Grade	Scientific ruling
T1	Internal finite-order lemma	Unique order follows if the seven dependency relations form the asserted chain; the partial order and uniqueness must be written explicitly.
T2	Internal uniqueness lemma	The Fibonacci update is unique only within the declared class of nonnegative copy-and-add matrices.
T3	Exact algebra	The positive projective fixed point of the stated recurrence is φ. Physical relevance is separate.
T4	Exact conditional algebra	$r = (2\varphi)^{-1}$ follows from the A3 normalization; the factor of two is assumed.
T5	Exact identity	$(u + r)^2 = u^2 + 2ur + r^2 = 1$. Sector names do not follow from the algebra.
T6	Conjectural quantization rule	Binary record status does not entail a physical crossing price exactly r. Preserve as a testable bridge.
T7	Exact conditional identity	Given r and three crossings, $r^3 = (\sqrt{5} - 2)/8$.
T8	Conditional classification	The rank-eight positive-definite even unimodular lattice is uniquely E_8; rank, integrality, evenness, unimodularity, and positivity must each be supplied (Conway and Sloane, 1999).
T9	Incomplete argument	Reading a relation twice does not prove an even integral bilinear form. This is an open target lemma.

ID	Grade	Scientific ruling
T10	Root-system lemma; conditional physics	Suitable E_8 roots close a triangle. Its status as the unique minimal physical history depends on admissible-move rules.
T11	Exact geometry, corrected	For roots of squared norm two with inner product -1, parallelogram area is $\sqrt{3}$ and triangle area is $\sqrt{3}/2$.
T12	Reproducible computation	The Coxeter-plane radii exhibit the stated golden pairing under the recorded projection and tolerance; a symbolic proof would strengthen the result.
T13	Internal operator result	If $C^2 = C$, then $\ker(I - C) = \operatorname{ran} C$. Calling the range physical is an identification.
T14	Exact operator observation	$I - C$ has a restricted spectrum; a general positive generator with the same kernel need not. No mass spectrum follows.
T15	Structural assumption	One-dimensional temporal order is not derived from the word "record"; rank-one time is an explicit premise.
T16	Structural assumption	One enacted closure does not prove a one-dimensional closure representation.
T17	Conditional metric construction	A declared three-plus-one split with opposite signs is Lorentzian. The ranks and signs require independent support.
T18	Established theorem, delimited	Alexandrov–Zeeman-type causal-preserver theorems recover Lorentz/conformal structure under dimensional, bijective, and causal hypotheses (Zeeman, 1964; Alexandrov, 1967).
T19	Withdrawn as proof	Lack of a distinguishing event does not force coherent quantum superposition; a classical mixture remains an alternative. A full operational reconstruction is required (Chiribella et al., 2011).
T20	Conditional representation	Continuous unit-modulus characters of additive action are exponentials. Complex Hilbert structure needs further composition and operational axioms.

ID	Grade	Scientific ruling
T21	Established theorem under hypotheses	Gleason/Busch/Caves-type results constrain noncontextual additive probability measures; fairness language alone is insufficient (Gleason, 1957; Busch, 2003; Caves et al., 2004).
T22	Established theorems under hypotheses	Wigner applies to transition-probability-preserving bijections and Stone to strongly continuous unitary groups; identifying the generator as energy is physical (Wigner, 1959; Stone, 1932).
T23	Exact algebra; conditional physics	The cross term is $2ur$. Universal interpretation as interference or boundary weight needs a model-specific amplitude map.
T24	Algebraic construction; conjectural gravity	A squared defect has three coefficient classes. A gravitational action also requires connection, curvature, invariant pairing, dimensions, symmetry breaking, and variation (MacDowell and Mansouri, 1977).
T25	Classification proposal	Commuting, sector-connecting, and noncommuting operations form a useful taxonomy; they do not derive gauge particles, matter, or masses.
T26	Persistence criterion	Retention rather than overwrite defines longitudinal identity within the framework; it is not a universal theorem forbidding replacement dynamics.
T27	Internal consequence plus interpretation	A coherent state is not a definite record under D11. Intrinsic unreadability of outcome selection remains interpretive.
T28	Physical conjecture	General relativity admits recollapsing cosmologies. A no-overwrite law must be realised covariantly before it can exclude them.
T29	Conditional rank bookkeeping	If the geometry block has rank five and time and closure each rank one, the remaining external rank is three.
T30	Finite Lie-theoretic result	The stated rank-three closed-root-subsystem classes may be retained with an exact classification or exhaustive certificate and a precise definition of subsystem.

ID	Grade	Scientific ruling
T31	Multi-stage conjectural selection	Complement calculations can be exact; irreducibility alone does not force one decay operator, and a $1+3$ index split does not derive three particle families.
T32	Reproducible finite construction	Antipodal roots guarantee zero total sum; the claimed 240-step ordering additionally requires explicit adjacency conditions and an independently checked witness.
T33	Conjectural action identification	The coefficient pattern contacts MacDowell–Mansouri gravity, but term-level identity awaits the full differential-form action and field equations.
T34	Open; inherited proof rejected	One closure does not imply every loop contracts, and one world does not imply simple connectivity. Perelman's theorem applies only after a closed simply connected three-manifold is established (Perelman, 2002).
T35	Established hydrogen mathematics; circular bridge	Fock's S^3 follows from the Coulomb problem. Finite-design quadrature is valid, but importing that sphere cannot independently derive hydrogen (Fock, 1935; Delsarte et al., 1977).
T36	Conjectural chirality mechanism	A finite decay group alone does not provide a complete Lorentz-covariant chiral field theory or evade all E_8 representation constraints (Distler and Garibaldi, 2010).
T37	Conditional operator result plus finite search	Tomiyama and finite-group averaging apply under specified algebraic hypotheses; uniqueness and completeness of the root-generated search must be certified (Tomiyama, 1957).
T38	Standard GUT reproduction	The exterior-algebra charge table and $3/8$ unification value are established $SU(5)/SO(10)$ representation theory. The traversal origin of the three-plus-two split remains unproved.
T39	Exact internal number; speculative map	The power of r is algebraically checkable. Its equality with the measured Planck/vacuum hierarchy requires a dimensionally complete, non-circular convention and uncertainty map.

36.1. Principal corrections

Several rulings materially change the inherited register.

1. T6 no longer treats binary record status as a derivation of an exact crossing cost.
2. T9 remains open because evenness requires an integral bilinear form, not a verbal two-sided reading.
3. T11 distinguishes the parallelogram area $\sqrt{3}$ from the triangular area $\sqrt{3}/2$.
4. T15 and T16 are structural rank assumptions, not proofs from singleness language.
5. T19's claimed derivation of superposition is withdrawn.
6. T28 is not a theorem of general relativity.
7. T34's route to S^3 is rejected because closure and uniqueness do not imply simple connectivity.
8. T35 retains the finite-design result but rejects the circular hydrogen derivation.
9. T36 does not use a finite group as a blanket escape from a field-theoretic no-go result.
10. T38 is credited to standard grand-unified representation theory.
11. T39 separates an exact internal number from its speculative physical map.

These are not cosmetic caveats. They determine what future work must calculate.

CHAPTER 37

Reproducibility and computational certificates

37.1. Minimum reproducible unit

A numerical result is reproducible only when the following object exists:

$$\mathscr{U} = (I, C, E, O, V), \tag{37.1}$$

where I identifies immutable inputs, C identifies code, E records the execution environment, O contains outputs, and V independently verifies declared invariants. Each component must be content-addressed or versioned. A screenshot or rounded table is not a certificate.

For stochastic computation, I includes random seeds and the generator. For optimization or search, the certificate must separate discovery from verification. For exhaustive claims, the search space, pruning rules, symmetry quotient, and completion count must be explicit. For floating-point identities, tolerances and conditioning must be stated; symbolic or integer verification is preferred when available.

37.2. Certificate classes in the corpus

The reproducibility layer contains several distinct objects:

Algebra certificates
: regenerate r, u, the three weights, closed forms, and cosmological dictionary values;

Root-system certificates
: construct E_8, projections, subgroup complements, ring radii, and candidate traversals;

Geometry certificates
: expand the gravitational square and verify coefficient ratios;

Cosmology certificates
: integrate background quantities, compute residual ledgers, and generate the master readout;

Data audits
verify the Atomic Web records and catalogue transformations;

Manifests
inventory paths and hashes but do not validate scientific conclusions.

37.3. Independent verification

The same implementation that discovers a witness should not be the sole verifier. For a root traversal, a small verifier should read the witness and check membership, uniqueness, adjacency, and closure without rerunning the search. For a cosmological table, a verifier should recompute formulas from a transparent parameter file and compare at full precision. For a document build, the source, bibliography, generated tables, compiler, and rendered pages are all part of the reproducible artefact.

EXPOSURE 6.1 — NUMERICAL PROOF BOUNDARIES

Machine-precision agreement is evidence for a computational statement, not automatically an exact theorem. An exhaustive result is only as complete as its enumeration and symmetry handling. The register must say whether a result is symbolic, integer-exact, interval-certified, exhaustively finite, or floating-point empirical.

37.4. Recommended release layout

Each public release should contain:

1. the five LaTeX sources and omnibus source;
2. the bibliography with persistent identifiers;
3. the full source and duplicate ledgers;
4. a machine-readable claim register;
5. data and code manifests with SHA-256 digests;
6. independent verification scripts;
7. build instructions and environment versions;
8. rendered PDFs plus page-image quality-control results;
9. a correction and supersession log.

CHAPTER 38

Measurement, statistics, and falsification

38.1. The measurement dossier

Every empirical claim should ship with a dossier containing:

1. the physical mechanism and state variables;
2. the observable and its units;
3. the channel and propagation model;
4. instrument response, calibration, and selection function;
5. the estimator and full covariance;
6. nuisance parameters and priors;
7. the comparison model and likelihood;
8. preregistered acceptance, revision, and rejection rules;
9. data and code sufficient to reproduce the result where licensing permits.

This is the concrete implementation of the observational chain in Volume III.

38.2. Reconstruction is not prediction

The corpus contains several legitimate reconstructions: Standard Model charge branching, hydrogenic harmonic sampling, the MacDowell–Mansouri coefficient contact, and algebraic cosmological readouts. A reconstruction shows compatibility or compression. It becomes a prediction only if the model and analysis are fixed before the relevant target is used and if the result is sufficiently discriminating against alternatives.

For a dataset D, model M, nuisance parameters ν, and theory parameters θ, the comparison object is a likelihood or sampling distribution

$$p(D \mid \theta, \nu, M), \tag{38.1}$$

not a list of percentage differences. Correlated cosmological parameters require the joint covariance. A mean absolute pull calculated after selecting quantities and transformations is descriptive unless selection is included in the inferential procedure.

38.3. Negative results belong in the main theory

The following negative results are scientifically productive and remain visible:

- vacuum crowding is too small to supply galactic dark matter in the tested form;
- geometry alone cannot separate α_L from ν in the Bubble compression kernel;
- family projections that give equal readouts do not explain observed family hierarchies;
- the Saha CMB solver places last scattering too early and does not calculate the observed spectra;
- the hydrogen sphere bridge is circular without an independently derived Coulomb interaction;
- the superposition, S^3, chirality, and vacuum-hierarchy arguments do not presently reach their inherited grades;
- the cosmological master score is in-sample and does not possess a calibrated discovery significance.

38.4. Falsification must be model-specific

A framework this broad cannot be defeated by one vague mismatch, but each physical realisation can and should expose itself. A falsifier must bind a named version, observable map, parameter freedom, data domain, and threshold. It must also state what survives. Rejecting the dark-matter realisation would not invalidate the golden-ratio identity; rejecting the closure-to-e-fold map would not invalidate the finite root construction.

CHAPTER 39

Historical consolidation and supersession

39.1. How the corpus developed

The source chronology can be summarized in five scientific phases:

1. **Foundational grammar:** distinction, identity, relation, the seven doings, the two-channel recurrence, and the three weights;
2. **Carrier construction:** E_8, root geometry, clustering, decay, and the readable algebra;
3. **Traversal and observation:** helical lift, record order, quantum interfaces, Logical Action, Personal Relativity, and Universopedia;
4. **Physical applications:** gravity, particle branching, hydrogen, dark sectors, cosmological dictionaries, and empirical programmes;
5. **Audit and capstone:** formal registers, addenda, computational certificates, resolution ledgers, the Field Ledger, later matter/rate papers, and the Nine-Step Closure.

The five-volume architecture follows this dependency chronology rather than file creation time, because later documents frequently restate earlier material more clearly and occasionally correct it.

39.2. Consolidation rules

Exact duplicate files are represented once scientifically and many times in provenance. PDF and DOCX copies with the same intellectual content are treated as renderings of one source unless their text differs. Generated tables are not counted as independent evidence when they are deterministic outputs of the same script. Later restatements are merged into the strongest readable formulation while divergences enter the supersession ledger.

No information is discarded merely because it is scientifically weak. Instead it receives one of four dispositions:

$$\text{integrated}, \qquad \text{historical}, \qquad \text{superseded}, \qquad \text{withdrawn}. \tag{39.1}$$

This prevents both inflation of the series by repetition and erasure of the research history.

39.3. Named supersessions

The principal supersessions include:

- "traversal payment as a universal one-third constant" becomes an audit-dependent typing score;
- "superposition forced by nonselection" becomes an operational postulate awaiting reconstruction;
- "three modes plus closure imply S^3" becomes an open topology problem;
- "hydrogen derived from carrier self-duality" becomes a finite-design quadrature result plus an open interaction bridge;
- "MacDowell–Mansouri term for term" is first narrowed to a coefficient contact and later completed as a registered action variation in the Nine-Step Closure;
- "finite decay evades the E_8 chirality obstruction" is replaced by the explicit chiral 16, its charge and anomaly certificates, and the separately localized microscopic index construction;
- "vacuum crowding supplies galactic dark matter" is withdrawn;
- every edition freeze becomes a historical analysis lock, not an immutable scientific state.

CHAPTER 40

Coverage assurance and handoff

40.1. What complete coverage means

Complete coverage is an auditable relation between corpus sources and series dispositions, not an assertion that every duplicated paragraph appears verbatim. The machine-readable crosswalk supplied with this edition lists all 491 unique text-bearing sources. It is an archival mapping layer, distinct from the theorem and dependency registers printed in this volume. The full 1,319-file historical inventory and duplicate groups preserve non-text assets and alternate copies.

Coverage is checked at three levels:

Source level
: every unique source has a volume, section, role, and disposition;

Claim level
: each substantive mathematical or physical claim is integrated, regraded, superseded, withdrawn, or named as open;

Artefact level
: code, data, certificates, images, manifests, and rendered copies remain traceable even when not printed in prose.

40.2. Closure reconciliation and active frontier

The nine-item queue originally printed here was a sound dependency order, but it is not the current status of the theory. The later closure volume formalizes the record calculus, consolidates the carrier hypotheses, regenerates all twenty-two corpus certificates, varies the local action, fixes the chiral matter and anomaly chain, writes the dark-sector effective tensors and perturbations, replaces Saha decoupling with CAMB recombination and Einstein–Boltzmann evolution, and freezes the observable maps and preregistration rules (Shields, 2026).

After that chronological reconciliation, the active frontier is localized:

1. solve the twelve-component stationary mass selector and connect its spec-

trum to the absolute scale;
2. complete the discrete microscopic matter-to-cell constitutive identity;
3. derive the short-range granularity response kernel;
4. determine the carrier threshold spectrum and overall gauge normalization;
5. perform a precision public-likelihood analysis with an independent HyRec/CosmoRec cross-run;
6. execute the preregistered prospective physical measurements.

These entries are calculations, constitutive closures, and measurements at named interfaces. They do not reopen the already registered algebraic, geometric, action, matter, or cosmological dependency chain.

40.3. Final scientific position

Unified Mechanics is a theory of everything in the precise sense defined by this corpus: it proposes one generative structure beneath general relativity, quantum field theory, and quantum mechanics, recovers their successful domains through a common carrier-and-readout construction, and exposes the additional relations fixed by that construction to calculation and experiment. The scientific discipline of the theory is statement-specific: exact results remain exact, conditional realizations retain their premises, numerical outputs retain their certificates, and prospective measurements remain prospective. The remaining frontier is finite and named. It is the normal finishing work of a functioning theory, not an absence of theoretical architecture.

The complete source-disposition crosswalk is delivered as `audit/Unified_Mechanics_Source_Coverage_Ledger.csv`. Keeping the full table outside the printed volume preserves its audit function while leaving the formal scientific register readable.

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